XCPC Templates

Skyclad Observer

Continue

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Contents

1 代数结构 4
1.1 幺半群 4
1.2 幺半群作用 6
1.3 半群快速幂 7

2 模运算 8
2.1 Barrett 模乘 8
2.2 运算 8
2.3 模数类 9
2.4 all inverse 10

3 图论 10
3.1 Graph 类 10
3.2 线段树优化建图 12
3.3 最短路 13
3.3.1 Dijkstra 13
3.3.2 01-BFS 15
3.3.3 Bellman-Ford 17
3.3.4 SPFA 17
3.3.5 Floyd 18
3.3.6 路径重建 19
3.4 差分约束 19
3.5 拓扑排序 20
3.6 欧拉通路/回路 21
3.7 最小生成树 23
3.7.1 Kruskal 23
3.7.2 Prim 24
3.7.3 Borůvka 25
3.7.4 Kruskal 重构树 26
3.8 连通性 27
3.8.1 强连通分量 27
3.8.2 边双连通分量 28
3.8.3 圆方树 28
3.9 2-SAT 29
3.10 树上问题 30
3.10.1 Tree 类 30
3.10.2 树链剖分 32
3.10.3 树上启发式合并 34
3.10.4 点分治 35
3.10.5 LCA 36
3.10.6 树上倍增 38
3.10.7 虚树 39
3.10.8 树上阿贝尔群 39
3.10.9 树上幺半群 40

4 网络流 42
4.1 最大流最小割 42
4.2 最小费用最大流 45

5 数论 47
5.1 扩展欧几里得算法 47
5.2 线性筛 48

5.3 莫比乌斯反演 . . . 48
5.4 欧拉函数 . . . 49
5.5 Miller-Rabin . . . 51
5.6 Pollard-Rho . . . 51
5.7 约数构造 . . . 53

6 组合数学 . . . 53
6.1 组合数 . . . 53
6.2 康托展开 . . . 55
6.3 逆序对 . . . 56
6.4 容斥原理 . . . 56
6.5 zeta 变换 . . . 58
6.6 Polyomino . . . 59

7 数据结构 . . . 60
7.1 线段树 . . . 60
7.1.1 线段树 . . . 60
7.1.2 懒标记线段树 . . . 63
7.1.3 动态开点线段树 . . . 66
7.2 树状数组 . . . 70
7.2.1 树状数组 . . . 70
7.2.2 区间加、单点查树状数组 . . . 71
7.2.3 区间加、区间和树状数组 . . . 72
7.2.4 二维树状数组 . . . 72
7.3 PBDS . . . 74
7.3.1 哈希表 . . . 74
7.3.2 红黑树 . . . 74
7.3.3 配对堆 . . . 75
7.4 倍增 . . . 75
7.5 并查集 . . . 76
7.6 稀疏表 . . . 77
7.7 01-Trie . . . 78
7.8 笛卡尔树 . . . 79

8 线性代数 . . . 80
8.1 矩阵 . . . 80
8.2 高斯消元 . . . 82
8.3 GF(2) 空间 . . . 85
8.3.1 异或线性基 . . . 85
8.3.2 逆矩阵 . . . 86
8.3.3 转置 . . . 87
8.3.4 GF(2) 线性方程组 . . . 88

9 字符串 . . . 88
9.1 Manacher . . . 88
9.2 KMP . . . 89
9.3 Trie . . . 89
9.4 Z 函数 . . . 90
9.5 AC 自动机 . . . 91

10 随机化 . . . 92
10.1 随机数 . . . 92
10.2 随机数组 . . . 93
10.3 随机图 . . . 94

10.4 模拟退火 96

11 杂项 97
11.1 二分答案 97
11.2 三分 97
11.3 i128 98
11.4 日期 99
11.5 template 99
11.6 debug 100
11.7 对拍 100
11.8 对拍(with SPJ) 101
11.9 SPJ 101

1 代数结构

1.1 幺半群

```cpp
template <typename Mono>
struct RevMono {
    using Type = typename Mono::Type;
    static Type e() {
        return Mono::e();
    }
    static Type op(const Type& a, const Type& b) {
        return Mono::op(b, a);
    }
};

template <typename T>
struct AddMono {
    using Type = T;
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {a + b};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type inv(const Type& a) {
        return {-a};
    }
    static Type pow(const Type& a, i64 k) {
        return a * k;
    }
};

template <typename T>
struct MulMono {
    using Type = T;
    static Type e() {
        return {1};
    }
    static Type op(const Type& a, const Type& b) {
        return {a * b};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type inv(const Type& a) {
        return T{1} / a;
    }
};

```

```cpp
template <typename T>
struct MinMono {
    using Type = T;
    static Type e() {
        return {inf<T>};
    }
    static Type op(const Type& a, const Type& b) {
        return {min(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct MaxMono {
    using Type = T;
    static Type e() {
        return {-inf<T>};
    }
    static Type op(const Type& a, const Type& b) {
        return {max(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct GcdMono {
    using Type = T;
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {gcd(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
```

```cpp
struct LcmMono {
    using Type = T;
    static Type e() {
        return {1};
    }
    static Type op(const Type& a, const Type& b) {
        return {lcm(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct XorMono {
    using Type = T;
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {a ^ b};
    }
    static Type inv(const Type& a) {
        return {a};
    }
    static Type power(const Type& a, i64 k) {
        return (k & 1) ? a : e();
    }
};
```

1.2 幺半群作用

```cpp
// Template
// struct ActMono {
//     using Mono = Mono;
//     using Tag = Tag;
//     using X = Mono::Type;
//     using Y = Tag::Type;
//     static X act(const X& x, const Y& y, int len) {
//         return {};
//     }
// };

template <typename T>
struct SumAdd {
    using Mono = AddMono<T>;
    using Tag = AddMono<T>;
```

```
    using X =  Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y * len};
    }
};

template <typename T>
struct SumMul {
    using Mono = AddMono<T>;
    using Tag = MulMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x * y};
    }
};

template <typename T>
struct MinAdd {
    using Mono = MinMono<T>;
    using Tag = AddMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y};
    }
};

template <typename T>
struct MaxAdd {
    using Mono = MaxMono<T>;
    using Tag = AddMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y};
    }
};
```

1.3 半群快速幂

```
template <typename Mono>
typename Mono::Type power(typename Mono::Type a, i64 b) {
    if constexpr (requires { Mono::pow(a, b); }) {
        return Mono::pow(a, b);
    } else {
        auto res = Mono::e();
        while (b > 0) {
            if (b & 1) {
```

```
                res = Mono::op(res, a);
            }
            a = Mono::op(a, a);
            b >>= 1;
        }
        return res;
    }
}
```

2 模运算

2.1 Barrett 模乘

```
class Barrett {
public:
    constexpr Barrett(uint32_t Mod) : mod_(Mod), mu_(static_cast<uint64_t>(-1) / Mod + 1) {}

    constexpr uint32_t mod() const {
        return mod_;
    }

    constexpr uint32_t multiply(uint32_t a, uint32_t b) const {
        uint64_t z = static_cast<uint64_t>(a) * b;
        uint64_t q = static_cast<uint64_t>((static_cast<__uint128_t>(z) * mu_) >> 64);
        uint64_t y = q * mod_;
        return static_cast<uint32_t>(z - y + (z < y ? mod_ : 0));
    }

private:
    uint32_t mod_;
    uint64_t mu_;
};
```

2.2 运算

```
i64 mul(i64 a, i64 b, i64 m) {
    return static_cast<i128>(a) * b % m;
}

i64 power(i64 a, i64 b, i64 m) {
    i64 res = 1 % m;
    while (b > 0) {
        if (b & 1) {
            res = mul(res, a, m);
        }
        a = mul(a, a, m);
        b >>= 1;
    }
```

```
    return res;
}

template <typename T, std::integral U>
constexpr T power(T a, U b) {
    assert(b >= 0);
    T res{1};
    while (b) {
        if (b & 1) {
            res *= a;
        }
        b >>= 1;
        a *= a;
    }
    return res;
}
```

2.3 模数类

```
template <u32 P>
struct Mint {
    u32 v;
    Mint(i64 v_ = 0) : v((v_ %= P) < 0 ? v_ + P : v_) {}
    Mint inv() const {
        i64 x, y;
        exgcd(v, P, x, y);
        return x;
    }
    Mint& operator+=(Mint rhs) & {
        v += rhs.v;
        if (v >= P) v -= P;
        return *this;
    }
    Mint& operator-=(Mint rhs) & {
        v -= rhs.v;
        if (v >= P) v += P;
        return *this;
    }
    Mint& operator*=(Mint rhs) & {
        v = u64(v) * rhs.v % P;
        return *this;
    }
    Mint& operator/=(Mint rhs) & {
        return *this *= rhs.inv();
    }
    friend Mint operator-(Mint rhs) { return rhs.v ? Mint(P - rhs.v) : rhs; }
    friend Mint operator+(Mint lhs, Mint rhs) { return lhs += rhs; }
    friend Mint operator-(Mint lhs, Mint rhs) { return lhs -= rhs; }
    friend Mint operator*(Mint lhs, Mint rhs) { return lhs *= rhs; }
    friend Mint operator/(Mint lhs, Mint rhs) { return lhs /= rhs; }
```

```
    friend bool operator==(Mint lhs, Mint rhs) { return lhs.v == rhs.v; }
    friend bool operator!=(Mint lhs, Mint rhs) { return lhs.v != rhs.v; }
    friend istream& operator>>(istream& is, Mint& v) { i64 x; is >> x; v = x; return is; }
    friend ostream& operator<<(ostream& os, const Mint& v) { return os << v.v; }
};

constexpr u32 P = 998244353;
using Fp = Mint<P>;
```

2.4 all inverse

```
template <typename Fp>
vector<Fp> allInverse(const vector<Fp>& a) {
    for (auto& v : a) {
        assert(v != 0);
    }
    int n = a.size();
    vector<Fp> res(n + 1);
    res[0] = Fp{1};
    for (int i = 0; i < n; i++) {
        res[i + 1] = res[i] * a[i];
    }
    Fp inv = res[n].inverse();
    res.pop_back();
    for (int i = n - 1; i >= 0; i--) {
        res[i] *= inv;
        inv *= a[i];
    }
    return res;
}
```

3 图论

3.1 Graph 类

```
template <typename T>
struct Edge {
    int u, v;
    T w;
};

template <typename T = int, bool dir = true>
struct Graph {
    int n;
    vector<Edge<T>> edges;
    vector<vector<pair<int, int>>> adj;

    Graph() : n(0) {}
```

```
14      Graph(int n_) : n(n_), adj(n_) {}
15
16      int add(int u, int v, T w = {1}) {
17          int id = edges.size();
18          edges.emplace_back(u, v, w);
19          adj[u].emplace_back(v, id);
20          if constexpr (!dir) {
21              adj[v].emplace_back(u, id);
22          }
23          return id;
24      }
25
26      auto& operator[](int u) const {
27          return adj[u];
28      }
29  };
30
31  template <typename T>
32  using Digraph = Graph<T, true>;
33  template <typename T>
34  using Undigraph = Graph<T, false>;
35
36  template <typename T>
37  Digraph<T> reverse(const Digraph<T>& g) {
38      Digraph<T> rg(g.n);
39      for (auto& [u, v, w] : g.edges) {
40          rg.add(v, u, w);
41      }
42      return rg;
43  }
44
45  template <typename T>
46  vector<int> degree(const Undigraph<T>& g) {
47      vector<int> deg(g.n);
48      for (auto& e : g.edges) {
49          deg[e.u]++;
50          deg[e.v]++;
51      }
52      return deg;
53  }
54
55  template <typename T>
56  pair<vector<int>, vector<int>> degree(const Digraph<T>& g) {
57      vector<int> in(g.n), out(g.n);
58      for (auto& e : g.edges) {
59          in[e.v]++;
60          out[e.u]++;
61      }
62      return {in, out};
63  }
64
65  template <typename T, bool dir>
```

```cpp
Graph<T, dir> induced(const Graph<T, dir>& g, const vector<int>& a) {
    int n = a.size();
    vector<int> id(g.n, -1);
    for (int i = 0; i < n; i++) {
        id[a[i]] = i;
    }
    Graph<T, dir> res(n);
    for (auto& e : g.edges) {
        if (id[e.u] != -1 && id[e.v] != -1) {
            res.add(id[e.u], id[e.v], e.w);
        }
    }
    return res;
}

template <typename T>
Undigraph<T> toUndigraph(const Digraph<T>& g) {
    Undigraph<T> res(g.n);
    for (auto& e : g.edges) {
        res.add(e.u, e.v, e.w);
    }
    return res;
}
```

3.2 线段树优化建图

```cpp
template <typename T>
struct RangeGraph {
    int n, tot;
    vector<Edge<T>> edges;

    RangeGraph() {}
    RangeGraph(int n_) : n(n_), tot(3 * n) {
        for (int i = 2; i < 2 * n; i++) {
            add(out(i / 2), out(i), T{});
            add(in(i), in(i / 2), T{});
        }
    }

    int out(int i) {
        return (i >= n ? i - n : n + i);
    }
    int in(int i) {
        return (i >= n ? i - n : 2 * n + i);
    }

    int add(int u, int v, T w = {1}) {
        int id = edges.size();
        edges.emplace_back(u, v, w);
        return id;
```

```
    }

    void toRange(int u, int l, int r, T w = {1}) {
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) {
                add(u, out(l++), w);
            }
            if (r & 1) {
                add(u, out(--r), w);
            }
        }
    }

    void fromRange(int l, int r, int v, T w = {1}) {
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) {
                add(in(l++), v, w);
            }
            if (r & 1) {
                add(in(--r), v, w);
            }
        }
    }

    void RangeToRange(int l1, int r1, int l2, int r2, T w = {1}) {
        int u = tot++;
        fromRange(l1, r1, u, w);
        toRange(u, l2, r2, T{});
    }

    Digraph<T> build() {
        Digraph<T> g(tot);
        for (auto& [u, v, w] : edges) {
            g.add(u, v, w);
        }
        return g;
    }
};
```

3.3 最短路

3.3.1 Dijkstra

```
template <typename T, bool dir>
pair<vector<T>, vector<int>> dijkstra(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    using P =  pair<T, int>;
    priority_queue<P, vector<P>, greater<P>> pq;

```

```
    dist[s] = 0;
    pq.emplace(0, s);

    while (!pq.empty()) {
        auto [d, u] = pq.top();
        pq.pop();
        if (d != dist[u]) {
            continue;
        }
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                pq.emplace(dist[v], v);
            }
        }
    }

    return {dist, pre};
}

template <typename T, bool dir>
pair<vector<T>, vector<int>> dijkstraDense(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    vector<int> done(n, false);

    dist[s] = 0;

    for (int _ = 0; _ < n; _++) {
        int u = -1;
        for (int v = 0; v < n; v++) {
            if (!done[v] && (u == -1 || dist[v] < dist[u])) {
                u = v;
            }
        }
        if (u == -1 || dist[u] == inf<T>) {
            break;
        }
        done[u] = true;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
            }
        }
    }

    return {dist, pre};
```

```cpp
}

template <typename T, bool dir>
tuple<vector<T>, vector<int>, vector<int>> dijkstra(const Graph<T, dir>& g, vector<int>& s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1), root(n, -1);
    using P =  pair<T, int>;
    priority_queue<P, vector<P>, greater<P>> pq;

    for (auto& u : s) {
        dist[u] = 0;
        root[u] = u;
        pq.emplace(0, u);
    }

    while (!pq.empty()) {
        auto [d, u] = pq.top();
        pq.pop();
        if (d != dist[u]) {
            continue;
        }
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                root[v] = root[u];
                pq.emplace(dist[v], v);
            }
        }
    }

    return {dist, pre, root};
}
```

3.3.2 01-BFS

```cpp
template <typename T, bool dir>
pair<vector<T>, vector<int>> bfs_01(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    deque<int> que;

    dist[s] = 0;
    que.push_front(s);

    while (!que.empty()) {
        auto u = que.front();
```

```
        que.pop_front();
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                if (w == 0) {
                    que.push_front(v);
                } else {
                    que.push_back(v);
                }
            }
        }
    }

    return {dist, pre};
}

template <typename T, bool dir>
tuple<vector<T>, vector<int>, vector<int>> bfs_01(const Graph<T, dir>& g, vector<int>& s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1), root(n, -1);
    deque<int> que;

    for (auto& u : s) {
        dist[u] = 0;
        root[u] = u;
        que.push_front(u);
    }

    while (!que.empty()) {
        auto u = que.front();
        que.pop_front();
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                root[v] = root[u];
                if (w == 0) {
                    que.push_front(v);
                } else {
                    que.push_back(v);
                }
            }
        }
    }

    return {dist, pre, root};
}
```

3.3.3 Bellman-Ford

```cpp
template <typename T, bool dir>
pair<vector<T>, vector<int>> bellmanFord(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);

    dist[s] = 0;

    for (int round = 1;; round++) {
        bool upd = false;
        for (int u = 0; u < n; u++) {
            if (dist[u] == inf<T>) {
                continue;
            }
            for (auto [v, id] : g[u]) {
                T w = g.edges[id].w;
                if (dist[v] > dist[u] + w) {
                    dist[v] = dist[u] + w;
                    pre[v] = u;
                    upd = true;
                    if (round >= n) {
                        return {{}, {}};
                    }
                }
            }
        }
        if (!upd) {
            break;
        }
    }

    return {dist, pre};
}

```

3.3.4 SPFA

```cpp
template <typename T, bool dir>
pair<vector<T>, vector<int>> spfa(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);

    dist[s] = 0;

    queue<int> q;
    vector<char> inq(n, false);
    vector<int> cnt(n, 0);
    q.push(s);
```

```
    inq[s] = true;

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        inq[u] = false;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                if (!inq[v]) {
                    q.push(v);
                    inq[v] = true;
                    cnt[v]++;
                    if (cnt[v] >= n) {
                        return {{}, {}};
                    }
                }
            }
        }
    }

    return {dist, pre};
}
```

3.3.5 Floyd

```
template <typename T>
void floyd(vector<vector<T>>& dist) {
    int n = dist.size();
    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (dist[i][k] < inf<T> && dist[k][j] < inf<T>) {
                    dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
                }
            }
        }
    }
}

template <typename T, bool dir>
vector<vector<T>> floyd(const Graph<T, dir>& g) {
    int n = g.n;
    vector dist(n, vector<T>(n, inf<T>));
    for (int u = 0; u < n; u++) {
        dist[u][u] = 0;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            dist[u][v] = min(dist[u][v], w);
```

```
        }
    }
    floyd(dist);
    return dist;
}

```

3.3.6 路径重建

```
vector<int> restorePath(vector<int>& pre, int t) {
    vector<int> path = {t};
    while (pre[path.back()] != -1) {
        path.push_back(pre[path.back()]);
    }
    reverse(path.begin(), path.end());
    return path;
}
```

3.4 差分约束

```
template <typename T>
struct DifferenceConstraints {
    int n, s;
    Digraph<T> graph;

    DifferenceConstraints() {}
    DifferenceConstraints(int n_) : n(n_), s(n), graph(n + 1) {
        for (int u = 0; u < n; u++) {
            graph.add(s, u);
        }
    }

    // x_v - x_u <= w
    void constraint(int u, int v, T w) {
        graph.add(u, v, w);
    }

    // x_v - x_u >= w
    void lowerBound(int u, int v, T w) {
        graph.add(v, u, -w);
    }

    // x_v - x_u = w
    void equality(int u, int v, T w) {
        constraint(u, v, w);
        lowerBound(u, v, w);
    }

```

```
    pair<bool, vector<T>> solve() {
        auto [dist, pre] = bellmanFord<T>(graph, s);
        if (dist.empty()) {
            return {false, {}};
        }
        dist.pop_back();
        return {true, dist};
    }
};
```

3.5 拓扑排序

```
template <typename T>
vector<int> toposort(const Digraph<T>& g) {
    int n = g.n;
    vector<int> topo, deg(n, 0);
    for (int u = 0; u < n; u++) {
        for (auto [v, id] : g[u]) {
            deg[v]++;
        }
    }
    for (int u = 0; u < n; u++) {
        if (deg[u] == 0) {
            topo.push_back(u);
        }
    }
    for (int i = 0; i < int(topo.size()); i++) {
        int u = topo[i];
        for (auto [v, id] : g[u]) {
            if (--deg[v] == 0) {
                topo.push_back(v);
            }
        }
    }
    return (int(topo.size()) != n ? vector<int>() : topo);
}

template <typename T>
vector<int> lexMinToposort(const Digraph<T>& g) {
    int n = g.n;
    vector<int> topo, deg(n, 0);
    for (int u = 0; u < n; u++) {
        for (auto [v, id] : g[u]) {
            deg[v]++;
        }
    }
    priority_queue<int, vector<int>, greater<int>> pq;
    for (int u = 0; u < n; u++) {
        if (deg[u] == 0) {
            pq.push(u);
```

```
        }
    }
    while (!pq.empty()) {
        int u = pq.top();
        pq.pop();
        topo.push_back(u);
        for (auto [v, id] : g[u]) {
            if (--deg[v] == 0) {
                pq.push(v);
            }
        }
    }
    return (int(topo.size()) != n ? vector<int>() : topo);
}
```

3.6 欧拉通路/回路

给定起点 s，还需要满足所有有边的顶点属于同一个连通分量，且 s 不能是孤立点。

无向图：

- 奇度点的数量一定是偶数，因此不会出现只有 1 个奇度点的情况。
- 有 0 个奇度点时，存在从任意有边顶点 s 出发的欧拉回路。
- 有 2 个奇度点时，s 必须是其中一个，存在从 s 出发的欧拉通路。
- 奇度点超过 2 个时不存在欧拉通路。

有向图：

- 若所有顶点满足 $\text{in} = \text{out}$，存在从任意有边顶点 s 出发的欧拉回路。
- 若恰有一个顶点满足 $\text{out} = \text{in} + 1$，它必须是 s；同时恰有一个顶点满足 $\text{in} = \text{out} + 1$，存在从 s 出发的欧拉通路。
- 其他情况不存在欧拉通路。

```
// 0: no Eulerian walk, 1: Eulerian path, 2: Eulerian circuit
template <typename T, bool dir>
int isEulerian(const Graph<T, dir>& g, int s = -1) {
    int n = g.n;
    if (g.edges.empty()) {
        return 2;  // 无边：任意点都有一条长度 0 的平凡欧拉回路
    }
    vector<int> deg(n);
    DSU dsu(n);
    for (auto& e : g.edges) {
        dsu.merge(e.u, e.v);
        deg[e.u]++;
        deg[e.v] += (dir ? -1 : 1);
    }
    int root = g.edges[0].u;
    for (auto& e : g.edges) {
        if (!dsu.same(root, e.u)) {
            return 0;
        }
    }
    if (s != -1 && !dsu.same(root, s)) {
        return 0;
```

```
    }
    int diff = 0;
    for (int u = 0; u < n; u++) {
        diff += dir ? abs(deg[u]) : (deg[u] & 1);
    }
    if (diff == 0) {
        return 2;
    }
    if (diff != 2) {
        return 0;
    }
    if (s == -1) {
        return 1;
    }
    return (dir ? deg[s] == 1 : (deg[s] & 1));
}

template <typename T, bool dir>
bool isEulerianPath(const Graph<T, dir>& g, int s = -1) {
    return isEulerian(g, s) >= 1;
}
template <typename T, bool dir>
bool isEulerianCircuit(const Graph<T, dir>& g, int s = -1) {
    return isEulerian(g, s) == 2;
}

template <typename T, bool dir>
pair<vector<int>, vector<int>> findEulerian(const Graph<T, dir>& g, int s = -1) {
    int n = g.n;
    int m = g.edges.size();
    assert(n > 0);
    assert(s == -1 || (0 <= s && s < n));
    if (s == -1) {
        vector<int> deg(n);
        for (auto& e : g.edges) {
            deg[e.u]++;
            deg[e.v] += (dir ? -1 : 1);
        }
        if constexpr (dir) {
            s = max_element(deg.begin(), deg.end()) - deg.begin();
            if (deg[s] == 0) {
                s = m == 0 ? 0 : g.edges[0].u;
            }
        } else {
            s = (m == 0 ? 0 : g.edges[0].u);
            for (int u = 0; u < n; u++) {
                if (deg[u] & 1) {
                    s = u;
                    break;
                }
            }
        }
```

```
    }
    if (m == 0) {
        return {{s}, {}};
    }
    vector<int> bal(n), ptr(n), vs, es;
    vector<char> used(m, false);
    vector<pair<int, int>> stk = {{s, -1}};
    bal[s]++;
    while (!stk.empty()) {
        auto [u, eid] = stk.back();
        if (ptr[u] == int(g.adj[u].size())) {
            vs.push_back(u);
            if (eid != -1) {
                es.push_back(eid);
            }
            stk.pop_back();
        } else {
            auto [v, id] = g.adj[u][ptr[u]++];
            if (used[id]) {
                continue;
            }
            bal[u]--;
            bal[v]++;
            used[id] = true;
            stk.emplace_back(v, id);
        }
    }
    for (auto& x : bal) {
        if (x < 0) {
            return {{}, {}};
        }
    }
    if (int(vs.size()) != m + 1) {
        return {{}, {}};
    }
    reverse(vs.begin(), vs.end());
    reverse(es.begin(), es.end());
    return {vs, es};
}
```

3.7 最小生成树

3.7.1 Kruskal

```
template <typename T>
pair<vector<int>, T> kruskal(const Undigraph<T>& g) {
    vector<int> ord(g.edges.size());
    iota(ord.begin(), ord.end(), 0);
    sort(ord.begin(), ord.end(), [&](int i, int j) {
        return g.edges[i].w < g.edges[j].w;
    });
```

```

    T cost = 0;
    vector<int> res;
    DSU dsu(g.n);
    for (auto& id : ord) {
        auto& e = g.edges[id];
        if (dsu.merge(e.u, e.v)) {
            res.push_back(id);
            cost += e.w;
        }
    }
    return {res, cost};
}
```

3.7.2 Prim

```
template <typename T, typename F>
vector<Edge<T>> prim(int n, F&& cost) {
    vector<Edge<T>> res;
    vector<T> wei(n, inf<T>);
    vector<int> pre(n);

    auto add = [&](int u) -> void {
        for (int v = 0; v < n; v++) {
            if (pre[v] == -1) {
                continue;
            }
            T w = cost(u, v);
            if (wei[v] > w) {
                wei[v] = w;
                pre[v] = u;
            }
        }
        wei[u] = inf<T>;
        pre[u] = -1;
    };

    add(0);
    for (int i = 0; i < n - 1; i++) {
        int u = min_element(wei.begin(), wei.end()) - wei.begin();
        res.emplace_back(pre[u], u, wei[u]);
        add(u);
    }

    return res;
}

template <typename T>
pair<vector<int>, T> prim(Undigraph<T>& g) {
    int n = g.n;
```

```
    vector<int> res, pre(n, -2);
    vector<T> wei(n, inf<T>);
    T cost = 0;
    auto add = [&](int u) -> void {
        for (auto& [v, id] : g[u]) {
            if (pre[v] == -1) {
                continue;
            }
            T w = g.edges[id].w;
            if (wei[v] > w) {
                wei[v] = w;
                pre[v] = id;
            }
        }
        wei[u] = inf<T>;
        pre[u] = -1;
    };
    add(0);
    for (int i = 0; i < n - 1; i++) {
        int u = min_element(wei.begin(), wei.end()) - wei.begin();
        if (wei[u] == inf<T>) {
            break;
        }
        res.push_back(pre[u]);
        cost += wei[u];
        add(u);
    }
    return {res, cost};
}
```

3.7.3 Borůvka

```
template <typename T>
pair<vector<int>, T> boruvka(Undigraph<T>& g) {
    auto& edges = g.edges;
    int n = g.n;
    T cost = 0;
    vector<int> res;
    DSU dsu(n);
    int cnt = n;

    while (cnt > 1) {
        vector<int> best(n, -1);
        int m = edges.size();

        for (int i = 0; i < m; i++) {
            auto& e = edges[i];
            int u = dsu.find(e.u);
            int v = dsu.find(e.v);
            if (u == v) {
                continue;
```

```
            }
            if (best[u] == -1 || e.w < edges[best[u]].w) {
                best[u] = i;
            }
            if (best[v] == -1 || e.w < edges[best[v]].w) {
                best[v] = i;
            }
        }

        bool upd = false;
        for (int u = 0; u < n; u++) {
            if (best[u] == -1) {
                continue;
            }
            auto& e = edges[best[u]];
            if (dsu.merge(e.u, e.v)) {
                res.push_back(best[u]);
                cost += e.w;
                cnt--;
                upd = true;
            }
        }
        if (!upd) {
            break;
        }
    }

    return {res, cost};
}
```

3.7.4 Kruskal 重构树

```
template <typename T>
pair<Undigraph<int>, vector<T>> kruskalTree(const Undigraph<T>& g) {
    int n = g.n;
    auto [ids, cost] = kruskal(g);
    int c = ids.size();

    Undigraph<int> res(n + c);
    vector<T> val(n + c);
    DSU dsu(n);
    vector<int> top(n);
    iota(top.begin(), top.end(), 0);

    for (int i = 0; i < c; i++) {
        int id = ids[i];
        int u = g.edges[id].u, v = g.edges[id].v;
        u = dsu.find(u), v = dsu.find(v);
        int p = n + i;
        res.add(p, top[u]);
```

```
        if (u != v) {
            res.add(p, top[v]);
        }
        dsu.merge(u, v);
        top[dsu.find(u)] = p;
        val[p] = g.edges[id].w;
    }

    return {res, val};
}
```

3.8 连通性

3.8.1 强连通分量

```
template <typename T>
pair<int, vector<int>> SCC(const Digraph<T>& g) {
    int n = g.n;
    vector<int> stk, dfn(n, -1), low(n), comp(n, -1);
    int cur = 0, cnt = 0;
    auto dfs = [&](auto&& self, int u) -> void {
        dfn[u] = low[u] = cur++;
        stk.push_back(u);
        for (auto& [v, id] : g[u]) {
            if (dfn[v] == -1) {
                self(self, v);
                low[u] = min(low[u], low[v]);
            } else if (comp[v] == -1) {
                low[u] = min(low[u], dfn[v]);
            }
        }
        if (dfn[u] == low[u]) {
            int y;
            do {
                y = stk.back();
                comp[y] = cnt;
                stk.pop_back();
            } while (y != u);
            cnt++;
        }
    };
    for (int i = 0; i < n; i++) {
        if (dfn[i] == -1) {
            dfs(dfs, i);
        }
    }
    for (auto& u : comp) {
        u = cnt - 1 - u;
    }
    return {cnt, comp};
```

```
}
```

3.8.2 边双连通分量

```cpp
template <typename T>
pair<int, vector<int>> EBCC(const Undigraph<T>& g) {
    int n = g.n;
    vector<int> stk;
    vector<int> dfn(n, -1), low(n), comp(n, -1);
    int cur = 0, cnt = 0;
    auto dfs = [&](auto&& self, int u, int parentEdge) -> void {
        dfn[u] = low[u] = cur++;
        stk.push_back(u);
        for (auto [v, id] : g[u]) {
            if (id == parentEdge) {
                continue;
            }
            if (dfn[v] == -1) {
                self(self, v, id);
                low[u] = min(low[u], low[v]);
            } else {
                low[u] = min(low[u], dfn[v]);
            }
        }
        if (dfn[u] == low[u]) {
            int v;
            do {
                v = stk.back();
                stk.pop_back();
                comp[v] = cnt;
            } while (v != u);
            cnt++;
        }
    };
    for (int u = 0; u < n; u++) {
        if (dfn[u] == -1) {
            dfs(dfs, u, -1);
        }
    }
    return {cnt, comp};
}
```

3.8.3 圆方树

```cpp
struct BlockCutTree {
    int n, tot;
    Undigraph<int> tree;

    BlockCutTree() {}
```

```
    template <typename T>
    BlockCutTree(const Undigraph<T>& g) : n(g.n), tree(n) {
        vector<int> dfn(n, -1), low(n), stk;
        int cur = 0;
        auto dfs = [&](auto&& self, int u) -> void {
            dfn[u] = low[u] = cur++;
            stk.push_back(u);
            for (auto [v, id] : g[u]) {
                if (dfn[v] == -1) {
                    self(self, v);
                    low[u] = min(low[u], low[v]);
                    if (low[v] >= dfn[u]) {
                        int b = tree.n++;
                        tree.adj.emplace_back();
                        tree.add(u, b);
                        int x;
                        do {
                            x = stk.back();
                            stk.pop_back();
                            tree.add(x, b);
                        } while (x != v);
                    }
                } else {
                    low[u] = min(low[u], dfn[v]);
                }
            }
        };
        for (int u = 0; u < n; u++) {
            if (dfn[u] == -1) {
                dfs(dfs, u);
                stk.pop_back();
            }
        }
        tot = tree.n;
    }

    bool isCut(int u) const {
        return tree[u].size() > 1;
    }
};
```

3.9 2-SAT

`addClause(u, f, v, g)` 加入一个析取子句：

$$(x_u = f) \vee (x_v = g).$$

这里 `x_i` 是第 `i` 个布尔变量，变量编号从 0 开始。`f` 和 `g` 决定文字的正负：`true` 表示正文字 x_i，`false` 表示负文字 $\neg x_i$。

例如：`addClause(3, true, 5, false)` 表示

$$x_3 \vee \neg x_5.$$

一个子句 $(A \vee B)$ 等价于两条蕴含边：

$$(\neg A) \to B, \quad (\neg B) \to A.$$

所以 addClause(u, f, v, g) 实际加入：

$$(x_u = \neg f) \to (x_v = g), \quad (x_v = \neg g) \to (x_u = f).$$

图中的编号约定是：$2i$ 表示变量 x_i 取假值，$2i+1$ 表示变量 x_i 取真值。 因此第一条边的起点是第一个文字的否定，终点是第二个文字本身；另一条边同理。

```cpp
struct TwoSat {
    int n;
    Digraph<int> graph;
    vector<int> ans;

    TwoSat(int n_) : n(n_), graph(2 * n), ans(n) {}

    // add clause: (x_u = f) ∨ (x_v = g)
    void addClause(int u, bool f, int v, bool g) {
        graph.add(2 * u + !f, 2 * v + g);
        graph.add(2 * v + !g, 2 * u + f);
    }

    // (x_u = f)
    void set(int u, bool f) {
        addClause(u, f, u, f);
    }

    // (x_u = f) -> (x_v = g)
    void implies(int u, int f, int v, int g) {
        addClause(u, !f, v, g);
    }

    pair<bool, vector<int>> satisfiable() {
        auto [cnt, id] = SCC(graph);
        for (int i = 0; i < n; i++) {
            if (id[2 * i] == id[2 * i + 1]) {
                return {false, {}};
            }
            ans[i] = id[2 * i] < id[2 * i + 1];
        }
        return {true, ans};
    }
};
```

3.10 树上问题

3.10.1 Tree 类

```cpp
template <typename T>
struct Tree {
    Undigraph<T>& g;
    int n, root;
    vector<int> fa, pe, dep, siz, in, out, seq;
```

```
    vector<T> dist;

    Tree(Undigraph<T>& g, int rt = 0) : g(g) {
        build(rt);
    }

    void build(int rt = 0) {
        n = g.n;
        root = rt;
        assert(0 <= root && root < n);
        assert(int(g.edges.size()) == n - 1);
        fa.assign(n, -1);
        pe.assign(n, -1);
        dep.assign(n, 0);
        siz.assign(n, 1);
        in.resize(n);
        out.resize(n);
        seq.resize(n);
        dist.assign(n, T{});

        int cur = 0;
        auto dfs = [&](auto&& self, int u) -> void {
            in[u] = cur;
            seq[cur++] = u;
            for (auto [v, id] : g[u]) {
                if (v == fa[u]) {
                    continue;
                }
                fa[v] = u;
                pe[v] = id;
                dep[v] = dep[u] + 1;
                dist[v] = dist[u] + g.edges[id].w;
                self(self, v);
                siz[u] += siz[v];
            }
            out[u] = cur;
        };
        dfs(dfs, root);
    }

    int eToV(int i) {
        auto& e = g.edges[i];
        return (dep[e.u] > dep[e.v] ? e.u : e.v);
    }

    int vToE(int u) {
        return pe[u];
    }

    vector<int> subtree(int u) {
        return vector<int>(seq.begin() + in[u], seq.begin() + out[u]);
    }
```

```
};
```

3.10.2 树链剖分

```
template <typename T>
struct HLD {
    Tree<T>& tree;
    vector<int> he, top;

    HLD(Tree<T>& tree) : tree(tree) {
        build();
    }

    void build() {
        int n = tree.n;
        he.assign(n, -1);
        top.resize(n);
        for (int v = 0; v < n; v++) {
            int u = tree.fa[v];
            if (u == -1) {
                continue;
            }
            if (he[u] == -1 || tree.siz[v] > tree.siz[he[u]]) {
                he[u] = v;
            }
        }
        int cur = 0;
        auto dfs = [&](auto&& self, int u, int t) -> void {
            top[u] = t;
            tree.in[u] = cur;
            tree.seq[cur++] = u;
            if (he[u] != -1) {
                self(self, he[u], t);
            }
            for (auto [v, id] : tree.g[u]) {
                if (v == tree.fa[u] || v == he[u]) {
                    continue;
                }
                self(self, v, v);
            }
            tree.out[u] = cur;
        };
        dfs(dfs, tree.root, tree.root);
    }

    int lca(int u, int v) {
        while (top[u] != top[v]) {
            if (tree.dep[top[u]] < tree.dep[top[v]]) {
                swap(u, v);
            }
```

```
47              u = tree.fa[top[u]];
48          }
49          return tree.dep[u] < tree.dep[v] ? u : v;
50      }
51
52      int dist(int u, int v) {
53          return tree.dep[u] + tree.dep[v] - 2 * tree.dep[lca(u, v)];
54      }
55
56      int jump(int u, int k) {
57          if (k > tree.dep[u]) {
58              return -1;
59          }
60          int d = tree.dep[u] - k;
61          while (tree.dep[top[u]] > d) {
62              u = tree.fa[top[u]];
63          }
64          return tree.seq[tree.in[u] - tree.dep[u] + d];
65      }
66
67      int jump(int u, int v, int k) {
68          int p = lca(u, v);
69          int a = tree.dep[u] - tree.dep[p];
70          int b = tree.dep[v] - tree.dep[p];
71          if (k > a + b) {
72              return -1;
73          }
74          if (k <= a) {
75              return jump(u, k);
76          }
77          return jump(v, a + b - k);
78      }
79
80      bool isAncester(int u, int v) {
81          return tree.in[u] <= tree.in[v] && tree.in[v] < tree.out[u];
82      }
83
84      int rootedParent(int root, int v) {
85          if (root == v) {
86              return v;
87          }
88          if (!isAncester(v, root)) {
89              return tree.fa[v];
90          }
91          return jump(root, tree.dep[root] - tree.dep[v] - 1);
92      }
93
94      int rootedSize(int root, int v) {
95          if (root == v) {
96              return tree.n;
97          }
98          if (!isAncester(v, root)) {
```

```
            return tree.siz[v];
        }
        return tree.n - tree.siz[rootedParent(root, v)];
    }

    int rootedLca(int root, int u, int v) {
        return lca(root, u) ^ lca(root, v) ^ lca(u, v);
    }

    template <typename F>
    void forPath(int u, int v, bool edge, F&& f) {
        vector<pair<int, int>> down;
        while (top[u] != top[v]) {
            if (tree.dep[top[u]] > tree.dep[top[v]]) {
                f(tree.in[top[u]], tree.in[u] + 1, true);
                u = tree.fa[top[u]];
            } else {
                down.emplace_back(tree.in[top[v]], tree.in[v] + 1);
                v = tree.fa[top[v]];
            }
        }
        bool rev = tree.dep[u] > tree.dep[v];
        int l = min(tree.in[u], tree.in[v]) + edge;
        int r = max(tree.in[u], tree.in[v]) + 1;
        if (l < r) {
            f(l, r, rev);
        }
        for (int i = int(down.size()) - 1; i >= 0; i--) {
            auto [l, r] = down[i];
            f(l, r, false);
        }
    }
};

```

3.10.3 树上启发式合并

适用于静态树上的离线子树统计，如不同颜色数、颜色频次、众数等。保留重子树的数据，暴力补入轻子树的数据。

- `add(u)`：加入节点 `u` 的数据，不应依赖加入顺序。
- `query(u)`：当前数据恰好包含 `u` 的整棵子树，每个节点一次；据此记录答案，不破坏统计状态。
- `reset()`：清空全部统计状态，建议记录修改过的位置，避免每次清空整个值域。

调用前统计状态须为空，结束后也为空。`add` 共调用 $O(n \log n)$ 次，`query` 调用 n 次；若单次加入、查询为 $O(1)$，且清空成本可摊到加入操作上，总时间为 $O(n \log n)$。

```
template <typename T, typename F1, typename F2, typename F3>
void dsuOnTree(const HLD<T>& hld, F1&& add, F2&& query, F3&& reset) {
    const auto& tree = hld.tree;
    for (int i = tree.n - 1; i >= 0; i--) {
        int u = tree.seq[i];
        add(u);
        for (auto& [v, id] : tree.g[u]) {
```

```
            if (v == tree.fa[u] || v == hld.he[u]) {
                continue;
            }
            for (int j = tree.in[v]; j < tree.out[v]; j++) {
                add(tree.seq[j]);
            }
        }
        query(u);
        if (hld.top[u] == u) {
            reset();
        }
    }
}
```

3.10.4 点分治

回调 `f(fa, vt, pos)`：`vt[i]` 为原树编号，`vt[0]` 为重心；`fa[i]` 为局部父亲编号，`fa[0] = -1`，其余满足 `0 <= fa[i] < i`。从 $b = 1$ 起，`[pos[b], pos[b+1])` 为各分支。只统计经过重心的路径，排除同分支点对；不应修改数组或保留引用。

当前块大小为 m，单次回调 $O(m)$ 时总时间 $O(n \log n)$，$O(m \log m)$ 时为 $O(n \log^2 n)$。初始化、清空也计入回调开销，避免每次扫描全局数组。

```
// f(fa, vt, pos)，局部编号 0 为重心，各分支占连续区间。
template <typename T, typename F>
void centroidDecompsition(const Tree<T>& tree, F&& f) {
    const auto& g = tree.g;
    vector<int> dead(tree.n), siz, fa, vt, pos;
    auto collect = [&](int u, int p) {
        int l = vt.size();
        vt.push_back(u);
        fa.push_back(p);
        for (int i = l; i < int(vt.size()); i++) {
            int u = vt[i];
            for (auto [v, id] : g[u]) {
                if (dead[v] || (fa[i] != -1 && v == vt[fa[i]])) {
                    continue;
                }
                vt.push_back(v);
                fa.push_back(i);
            }
        }
    };
    vector<int> rts{tree.root};
    for (int t = 0; t < int(rts.size()); t++) {
        vt.clear();
        fa.clear();
        collect(rts[t], -1);
        int m = vt.size(), c = -1;
        siz.assign(m, 1);
        for (int i = m - 1; i >= 0; i--) {
            if (siz[i] >= (m + 1) / 2) {
```

```
                c = vt[i];
                break;
            }
            siz[fa[i]] += siz[i];
        }
        dead[c] = true;
        vt = {c};
        fa = {-1};
        pos = {0, 1};
        for (auto [v, id] : g[c]) {
            if (!dead[v]) {
                rts.push_back(v);
                collect(v, 0);
                pos.push_back(vt.size());
            }
        }
        f(fa, vt, pos);
    }
}
```

3.10.5 LCA

```
template <typename T>
struct LCA {
    Tree<T>& tree;
    int n, logn;
    vector<vector<int>> fa;

    LCA(Tree<T>& t) : tree(t), n(tree.n), logn(__lg(n)), fa(n, vector<int>(logn + 1)) {
        for (int u = 0; u < n; u++) {
            fa[u][0] = (tree.fa[u] == -1 ? u : tree.fa[u]);
        }
        for (int k = 1; k <= logn; k++) {
            for (int u = 0; u < n; u++) {
                fa[u][k] = fa[fa[u][k - 1]][k - 1];
            }
        }
    }

    int get(int u, int v) {
        if (tree.dep[u] < tree.dep[v]) {
            swap(u, v);
        }
        for (int i = logn; i >= 0; i--) {
            if (tree.dep[fa[u][i]] >= tree.dep[v]) {
                u = fa[u][i];
            }
        }
        if (u == v) {
            return u;
        }
```

```
        for (int i = logn; i >= 0; i--) {
            if (fa[u][i] != fa[v][i]) {
                u = fa[u][i];
                v = fa[v][i];
            }
        }
        return fa[u][0];
    }

    int dist(int u, int v) {
        int res = tree.dep[u] + tree.dep[v] - 2 * tree.dep[get(u, v)];
        return res;
    }
};

template <typename T>
struct FastLCA {
    struct DfnMin {
        struct Type {
            int dfn, fa;
        };
        static Type op(const Type& a, const Type& b) {
            return a.dfn < b.dfn ? a : b;
        }
    };

    Tree<T>& tree;
    int n;
    SparseTable<DfnMin> st;

    FastLCA(Tree<T>& tree) : tree(tree), n(tree.n) {
        vector<typename DfnMin::Type> seq(n);
        for (int u = 0; u < n; u++) {
            int p = tree.fa[u] == -1 ? u : tree.fa[u];
            seq[tree.in[u]] = {tree.in[p], p};
        }
        st = SparseTable<DfnMin>(seq);
    }

    int query(int u, int v) {
        if (u == v) {
            return u;
        }
        if (tree.in[u] > tree.in[v]) {
            swap(u, v);
        }
        return st.query(tree.in[u] + 1, tree.in[v] + 1).fa;
    }

    int dist(int u, int v) {
        return tree.dep[u] + tree.dep[v] - 2 * tree.dep[query(u, v)];
    }
```

```
};
```

3.10.6 树上倍增

```
template <typename Mono, typename T>
struct TreeLifting {
    using X = typename Mono::Type;

    Tree<T>& tree;
    BinaryLifting<Mono> lift;

    TreeLifting(Tree<T>& tree) : tree(tree), lift(tree.n) {
        int n = tree.n;
        for (int v = 0; v < n; v++) {
            if (v == tree.root) {
                lift.add(v, -1, Mono::e());
            } else {
                lift.add(v, tree.fa[v], Mono::from(tree.g.edges[tree.vToE(v)].w));
            }
        }
        lift.build();
    }

    int lca(int u, int v) {
        if (tree.dep[u] < tree.dep[v]) {
            swap(u, v);
        }
        u = lift.jump(u, tree.dep[u] - tree.dep[v]);
        if (u == v) {
            return u;
        }
        for (int k = lift.logn; k >= 0; k--) {
            if (lift.nxt[k][u] != lift.nxt[k][v]) {
                u = lift.nxt[k][u];
                v = lift.nxt[k][v];
            }
        }
        return lift.nxt[0][u];
    }

    pair<int, X> query(int u, int v) {
        int l = lca(u, v);
        X res = Mono::op(lift.query(u, tree.dep[u] - tree.dep[l]).second, lift.query(v, tree.dep[v] - tree.dep[l]).second);
        return {l, res};
    }
};
```

3.10.7 虚树

```cpp
template <typename T>
struct VirtualTree {
    TreeLifting<AddMono<i64>, T>& tree;
    vector<int> vt;
    vector<vector<pair<int, T>>> adj;

    VirtualTree(TreeLifting<AddMono<i64>, T>& t) : tree(t) {}

    pair<vector<int>&, vector<vector<pair<int, T>>>&> build(vector<int>& h) {
        vt = h;
        auto cmp = [&](int u, int v) { return tree.tree.in[u] < tree.tree.in[v]; };
        sort(vt.begin(), vt.end(), cmp);
        int m = vt.size();
        for (int i = 1; i < m; i++) {
            vt.push_back(tree.lca(vt[i - 1], vt[i]));
        }
        sort(vt.begin(), vt.end(), cmp);
        vt.erase(unique(vt.begin(), vt.end()), vt.end());
        int n = vt.size();
        adj.resize(n);
        for (auto& e : adj) {
            e.clear();
        }
        for (int i = 0; i < n - 1; i++) {
            int p = tree.lca(vt[i], vt[i + 1]);
            int u = lower_bound(vt.begin(), vt.end(), p, cmp) - vt.begin();
            int v = i + 1;
            T w = tree.tree.dep[vt[v]] - tree.tree.dep[p];
            adj[u].emplace_back(v, w);
            adj[v].emplace_back(u, w);
        }
        return {vt, adj};
    }
};
```

3.10.8 树上阿贝尔群

```cpp
template <typename T, typename Abel, bool edge = false>
struct TreeAbel {
    using X = typename Abel::Type;

    HLD<T>& hld;
    Tree<T>& tree;
    int n;
    Fenwick<Abel> sub;
    DualFenwick<Abel> path;

    TreeAbel(HLD<T>& hld_) : hld(hld_), tree(hld.tree), n(tree.n), sub(n), path(n) {}
    TreeAbel(HLD<T>& hld_, const vector<X>& init) : hld(hld_), tree(hld.tree), n(tree.n) {
```

```
        assert(int(init.size()) == n - edge);
        vector<X> a(n, Abel::e()), b(n, Abel::e());
        for (int i = 0; i < int(init.size()); i++) {
            int u = (edge ? tree.eToV(i) : i);
            a[tree.in[u]] = init[i];
        }
        for (int i = 0; i < n; i++) {
            int u = tree.seq[i], p = tree.fa[u];
            b[i] = a[i];
            if (p != -1) {
                b[i] = Abel::op(b[tree.in[p]], b[i]);
            }
        }
        sub = Fenwick<Abel>(a);
        path = DualFenwick<Abel>(b);
    }

    X rootQuery(int u) {
        return path.query(tree.in[u]);
    }

    X pathQuery(int u, int v) {
        int l = hld.lca(u, v);
        X x = rootQuery(l);
        X res = Abel::op(rootQuery(u), rootQuery(v));
        res = Abel::op(res, Abel::inv(Abel::op(x, x)));
        if constexpr (!edge) {
            res = Abel::op(res, sub.query(tree.in[l], tree.in[l] + 1));
        }
        return res;
    }

    X subtreeQuery(int u, int root = -1) {
        if (root == -1) {
            root = tree.root;
        }
        if (root == u) {
            return sub.query(n);
        }
        if (!hld.isAncester(u, root)) {
            return sub.query(tree.in[u] + edge, tree.out[u]);
        }
        int v = hld.jump(root, tree.dep[root] - tree.dep[u] - 1);
        return Abel::op(sub.query(0, tree.in[v]), sub.query(tree.out[v], n));
    }
};
```

3.10.9 树上幺半群

```
template <typename T, typename Mono, bool edge = false>
```

```
struct TreeMono {
    using X = typename Mono::Type;

    HLD<T>& hld;
    Tree<T>& tree;
    int n;
    SegmentTree<Mono> seg;
    SegmentTree<RevMono<Mono>> rseg;

    TreeMono(HLD<T>& hld_) : TreeMono(hld_, vector<X>(hld_.tree.n - edge, Mono::e())) {}
    TreeMono(HLD<T>& hld_, const vector<X>& init) : hld(hld_), tree(hld.tree), n(tree.n) {
        assert(int(init.size()) == n - edge);
        vector<X> a(n, Mono::e());
        for (int i = 0; i < int(init.size()); i++) {
            a[id(i)] = init[i];
        }
        seg = SegmentTree<Mono>(a);
        rseg = SegmentTree<RevMono<Mono>>(a);
    }

    void modify(int i, const X& x) {
        int p = id(i);
        seg.modify(p, x);
        rseg.modify(p, x);
    }

    void apply(int i, const X& x) {
        int p = id(i);
        modify(i, Mono::op(seg.query(p, p + 1), x));
    }

    X pathQuery(int u, int v) {
        X ans = Mono::e();
        hld.forPath(u, v, edge, [&](int l, int r, bool rev) {
            X cur = (rev ? rseg.query(l, r) : seg.query(l, r));
            ans = Mono::op(ans, cur);
        });
        return ans;
    }

    X subtreeQuery(int u, int root = -1) {
        if (root == -1) {
            root = tree.root;
        }
        if (root == u) {
            return seg.query(0, n);
        }
        if (!hld.isAncester(u, root)) {
            return seg.query(tree.in[u] + edge, tree.out[u]);
        }
        assert(!edge);
        int v = hld.jump(root, tree.dep[root] - tree.dep[u] - 1);
```

```
        return Mono::op(seg.query(0, tree.in[v]), seg.query(tree.out[v], n));
    }

    X allQuery() {
        return seg.query(0, n);
    }

    template <typename F>
    int maxPath(F&& pred, int u, int v) {
        X acc = Mono::e();
        if (edge && !pred(acc)) {
            return -1;
        }

        int ans = v;
        bool done = false;
        hld.forPath(u, v, edge, [&](int l, int r, bool rev) {
            if (done) {
                return;
            }
            X cur = rev ? rseg.query(l, r) : seg.query(l, r);
            X next = Mono::op(acc, cur);
            if (pred(next)) {
                acc = next;
                return;
            }
            int p = (rev ? rseg.minLeft(r, pred, acc) - 1 : seg.maxRight(l, pred, acc));
            int w = tree.seq[p];
            if constexpr (edge) {
                ans = (rev ? w : tree.fa[w]);
            } else {
                ans = (w == u ? -1 : hld.jump(w, u, 1));
            }
            done = true;
        });
        return ans;
    }

    int id(int i) {
        int u = (edge ? tree.eToV(i) : i);
        return tree.in[u];
    }
};
```

4 网络流

4.1 最大流最小割

```
template <typename T>
struct MaxFlow {
```

```
    struct Edge_ {
        int to;
        T cap;
        Edge_(int to_, T cap_) : to(to_), cap(cap_) {}
    };

    int n;
    vector<Edge_> e;
    vector<vector<int>> adj;
    vector<int> cur, level;

    MaxFlow() {}
    MaxFlow(int n_) : n(n_), adj(n), cur(n), level(n) {}

    bool bfs(int s, int t) {
        level.assign(n, -1);
        queue<int> q;
        level[s] = 0;
        q.push(s);
        while (!q.empty()) {
            const int u = q.front();
            q.pop();
            for (auto& i : adj[u]) {
                auto [v, c] = e[i];
                if (c > 0 && level[v] == -1) {
                    level[v] = level[u] + 1;
                    if (v == t) {
                        return true;
                    }
                    q.push(v);
                }
            }
        }
        return false;
    }

    T dfs(int u, int t, T f) {
        if (u == t) {
            return f;
        }
        auto r = f;
        for (int& i = cur[u]; i < int(adj[u].size()); i++) {
            const int& j = adj[u][i];
            auto [v, c] = e[j];
            if (c > 0 && level[v] == level[u] + 1) {
                auto a = dfs(v, t, min(r, c));
                e[j].cap -= a;
                e[j ^ 1].cap += a;
                r -= a;
                if (r == 0) {
                    return f;
                }
```

```
            }
        }
        return f - r;
    }

    void addEdge(int u, int v, T c) {
        adj[u].push_back(e.size());
        e.emplace_back(v, c);
        adj[v].push_back(e.size());
        e.emplace_back(u, 0);
    }

    T flow(int s, int t) {
        T ans = 0;
        while (bfs(s, t)) {
            cur.assign(n, 0);
            ans += dfs(s, t, numeric_limits<T>::max());
        }
        return ans;
    }

    vector<bool> minCut() {
        vector<bool> c(n);
        for (int i = 0; i < n; i++) {
            c[i] = (level[i] != -1);
        }
        return c;
    }

    struct Edge {
        int from;
        int to;
        T cap;
        T flow;
    };

    vector<Edge> edges() {
        vector<Edge> a;
        for (int i = 0; i < e.size(); i += 2) {
            Edge x;
            x.from = e[i + 1].to;
            x.to = e[i].to;
            x.cap = e[i].cap + e[i + 1].cap;
            x.flow = e[i + 1].cap;
            a.push_back(x);
        }
        return a;
    }
};
```

4.2 最小费用最大流

最小费用流的每次增广都要在残量网络上找最短路，而残量网络里有反向边、费用是负的，`Dijkstra` 直接用不了。势能数组 h 就是用来把边权「平移」成非负的。

做法是把每条边的费用换成约化费用

$$c'(u,v) = c(u,v) + h(u) - h(v),$$

并且每轮求完最短路后令 $h(v) \leftarrow h(v) + \mathrm{dist}'(v)$，其中 dist' 是用约化费用算出的最短路。这样之后的约化费用依然非负——由三角不等式 $\mathrm{dist}'(v) \le \mathrm{dist}'(u) + c'(u,v)$ 整理一下，正好就是新的 $c'(u,v) \ge 0$。

代价是路径长度被平移了常量 $h(s) - h(t)$，但它不影响谁是最短路。代码里每轮累加的是 $h(t)$：因为 $h(s)$ 恒为 0，更新后的 $h(t)$ 恰好等于原网络中 $s \to t$ 的真实最短费用。

初始势能默认全 0，这就要求**初始边权全部非负**；如果原图本来就有负费用边，得根据图的特性自行构建 h 或者直接先跑一遍 `SPFA` 得到初始 h，再从参数传进来。

```cpp
template <typename T>
struct MinCostFlow {
    struct Edge_ {
        int to;
        T cap;
        T cost;
        Edge_(int to_, T cap_, T cost_) : to(to_), cap(cap_), cost(cost_) {}
    };

    int n;
    vector<Edge_> e;
    vector<vector<int>> adj;
    vector<T> h, dist;
    vector<int> pre;

    bool dijkstra(int s, int t) {
        dist.assign(n, inf<T>);
        pre.assign(n, -1);
        priority_queue<pair<T, int>, vector<pair<T, int>>, greater<pair<T, int>>> que;
        dist[s] = 0;
        que.emplace(0, s);
        while (!que.empty()) {
            auto [d, u] = que.top();
            que.pop();
            if (dist[u] != d) {
                continue;
            }
            for (auto& i : adj[u]) {
                int v = e[i].to;
                if (e[i].cap > 0 && dist[v] > d + h[u] - h[v] + e[i].cost) {
                    dist[v] = d + h[u] - h[v] + e[i].cost;
                    pre[v] = i;
                    que.emplace(dist[v], v);
                }
            }
        }
        return dist[t] != inf<T>;
    }
```

```
39
40      MinCostFlow() {}
41      MinCostFlow(int n_) : n(n_), adj(n), h(n) {}
42
43      void addEdge(int u, int v, T cap, T cost) {
44          adj[u].push_back(e.size());
45          e.emplace_back(v, cap, cost);
46          adj[v].push_back(e.size());
47          e.emplace_back(u, 0, -cost);
48      }
49
50      void set(const vector<T>& potential) {
51          h = potential;
52      }
53
54      pair<T, T> flow(int s, int t, T need = inf<T>) {
55          T flow = 0;
56          T cost = 0;
57          while (flow < need && dijkstra(s, t)) {
58              for (int i = 0; i < n; i++) {
59                  if (dist[i] != inf<T>) {
60                      h[i] += dist[i];
61                  }
62              }
63              T aug = need - flow;
64              for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
65                  aug = min(aug, e[pre[i]].cap);
66              }
67              for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
68                  e[pre[i]].cap -= aug;
69                  e[pre[i] ^ 1].cap += aug;
70              }
71              flow += aug;
72              cost += aug * h[t];
73          }
74          return {flow, cost};
75      }
76
77      struct Edge {
78          int from;
79          int to;
80          T cap;
81          T cost;
82          T flow;
83      };
84
85      vector<Edge> edges() {
86          vector<Edge> a;
87          for (int i = 0; i < e.size(); i += 2) {
88              Edge x;
89              x.from = e[i + 1].to;
90              x.to = e[i].to;
```

```
            x.cap = e[i].cap + e[i + 1].cap;
            x.cost = e[i].cost;
            x.flow = e[i + 1].cap;
            a.push_back(x);
        }
        return a;
    }
};
```

5 数论

5.1 扩展欧几里得算法

```
i64 exgcd(i64 a, i64 b, i64& x, i64& y) {
    if (b == 0) {
        x = 1, y = 0;
        return a;
    }
    i64 d = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return d;
}

i64 inv(i64 a, i64 m) {
    i64 x, y;
    i64 d = exgcd(a, m, x, y);
    if (d != 1) {
        return -1;
    }
    return (x % m + m) % m;
}

bool solveLinear(i64 a, i64 b, i64 c, i64& x, i64& y) {
    i64 d = exgcd(a, b, x, y);
    if (c % d != 0) {
        return false;
    }
    i64 k = c / d;
    x *= k;
    y *= k;
    return true;
}

pair<i64, i64> solveMod(i64 a, i64 b, i64 m) {
    assert(m > 0);
    b *= -1;
    i64 x, y;
    i64 g = exgcd(a, m, x, y);
    if (g < 0) {
        g *= -1;
```

```
        x *= -1;
        y *= -1;
    }
    if (b % g != 0) {
        return {-1, -1};
    }
    x = x * (b / g) % (m / g);
    if (x < 0) {
        x += m / g;
    }
    return {x, m / g};
}

```

5.2 线性筛

```
vector<int> minp, primes;

void sieve(int n) {
    minp.assign(n + 1, 0);
    primes.clear();

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            primes.push_back(i);
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                break;
            }
        }
    }
}
```

5.3 莫比乌斯反演

莫比乌斯反演用来在「按约数求和」和「按倍数求和」之间互转。

$\mu(n)$ 的定义：$n=1$ 时为 1；n 是 k 个互异素数之积时为 $(-1)^k$；含平方因子时为 0。一切从这条恒等式出发：

$$\sum_{d|n}\mu(d)=[n=1].$$

于是有两种方向的反演：

$$f(n)=\sum_{d|n}g(d)\Longrightarrow g(n)=\sum_{d|n}\mu(d)f\left(\frac{n}{d}\right),$$

$$f(n)=\sum_{n|d}g(d)\Longrightarrow g(n)=\sum_{n|d}\mu\left(\frac{d}{n}\right)f(d).$$

实际做题更常用的是「换条件」的用法：把 $[\gcd(i,j)=1]$ 换成 $\sum_{d|\gcd(i,j)}\mu(d)$，再交换求和顺序，条件就从 gcd 挪到了 d：

$$\sum_{i=1}^{n}\sum_{j=1}^{m}[\gcd(i,j)=1]=\sum_{d=1}^{\min(n,m)}\mu(d)\left\lfloor\frac{n}{d}\right\rfloor\left\lfloor\frac{m}{d}\right\rfloor.$$

右边只要 μ 的前缀和就能整除分块，μ 用下面的 `sieve()` 线性筛出来。

```cpp
vector<int> minp, primes, mu;

void sieve(int n) {
    minp.assign(n + 1, 0);
    mu.assign(n + 1, 0);
    primes.clear();
    mu[1] = 1;

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            mu[i] = -1;
            primes.push_back(i);
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                mu[i * p] = 0;
                break;
            } else {
                mu[i * p] = -mu[i];
            }
        }
    }
}

```

5.4 欧拉函数

$$\varphi(n)=n\prod_{i=1}^{k}\left(1-\frac{1}{p_i}\right).$$

两条常用性质：它是积性的，即 $\gcd(a,b)=1$ 时 $\varphi(ab)=\varphi(a)\varphi(b)$；并且 $\sum_{d|n}\varphi(d)=n$，也即 $\varphi(n)=\sum_{d|n}\mu(d)\frac{n}{d}$。

欧拉定理：$\gcd(a,m)=1$ 时 $a^{\varphi(m)}\equiv 1(\bmod m)$。竞赛里更常用的是扩展欧拉定理，它给指数降幂—$b\geq\varphi(m)$ 时对任意 a 都有

$$a^b \equiv a^{b \bmod \varphi(m)+\varphi(m)} (\bmod m).$$

线性筛求 φ 的递推：素数的 $\varphi(p) = p-1$；对 $i \cdot p$，若 $p \mid i$ 则 $\varphi(ip) = p\varphi(i)$，否则 $\varphi(ip) = (p-1)\varphi(i)$。见下面的 `sieve()`；只求单个数用 `Phi()`，复杂度 $O(\sqrt{n})$。

```cpp
vector<int> minp, primes, phi;

// 线性筛（用于求解全部数的欧拉函数）
void sieve(int n) {
    minp.assign(n + 1, 0);
    phi.assign(n + 1, 0);
    primes.clear();
    phi[1] = 1;

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            primes.push_back(i);
            phi[i] = i - 1;
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                phi[i * p] = phi[i] * p;
                break;
            } else {
                phi[i * p] = phi[i] * (p - 1);
            }
        }
    }
}

// 求解单个数的欧拉函数
int Phi(int n) {
    int res = n;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0) {
                n /= i;
            }
            res = res / i * (i - 1);
        }
    }
    if (n > 1) {
        res = res / n * (n - 1);
    }
    return res;
}

```

5.5 Miller-Rabin

```cpp
bool isPrime(i64 n) {
    if (n < 2) {
        return false;
    }
    constexpr array<int, 10> smallPrimes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29};
    for (auto& p : smallPrimes) {
        if (n % p == 0) {
            return n == p;
        }
    }
    if (n < 31 * 31) {
        return true;
    }
    constexpr array<int, 7> bases = {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
    int s = __builtin_ctzll(n - 1);
    i64 d = (n - 1) >> s;
    for (auto& a : bases) {
        if (a % n == 0) {
            continue;
        }
        i64 cur = power(a, d, n);
        if (cur == 1) {
            continue;
        }
        bool witness = true;
        for (int r = 0; r < s; r++) {
            if (cur == n - 1) {
                witness = false;
                break;
            }
            cur = mul(cur, cur, n);
        }
        if (witness) {
            return false;
        }
    }
    return true;
}
```

5.6 Pollard-Rho

```cpp
i64 rho(i64 n, i64 c) {
    auto f = [&](i64 x) {
        return static_cast<i64>((static_cast<i128>(x) * x + c) % n);
    };
    i64 x = 1, y = 2, z = 1, q = 1;
    i64 g = 1;
    constexpr i64 m = 127;
    for (i64 r = 1; g == 1; r <<= 1) {
```

```
        x = y;
        for (i64 i = 0; i < r; i++) {
            y = f(y);
        }
        for (i64 k = 0; k < r && g == 1; k += m) {
            z = y;
            for (i64 i = 0; i < min(m, r - k); i++) {
                y = f(y);
                q = mul(q, abs(x - y), n);
            }
            g = gcd(q, n);
        }
    }
    if (g == n) {
        do {
            z = f(z);
            g = gcd(abs(x - z), n);
        } while (g == 1);
    }
    return g;
}

i64 primeFactor(i64 n) {
    assert(n >= 2);
    if (n % 2 == 0) {
        return 2;
    }
    while (!isPrime(n)) {
        n = rho(n, randInt(1, n));
    }
    return n;
}

vector<pair<i64, int>> factorize(i64 n) {
    vector<pair<i64, int>> pf;
    for (int p = 2; p < 100; p++) {
        if (p * p > n) {
            break;
        }
        if (n % p == 0) {
            int e = 0;
            do {
                n /= p;
                e++;
            } while (n % p == 0);
            pf.emplace_back(p, e);
        }
    }
    while (n > 1) {
        i64 p = primeFactor(n);
        int e = 0;
        do {
```

```
            n /= p;
            e++;
        } while (n % p == 0);
        pf.emplace_back(p, e);
    }
    sort(pf.begin(), pf.end());
    return pf;
}
```

5.7 约数构造

```
vector<i64> buildDivisors(const vector<pair<i64, int>>& factors) {
    vector<i64> divisors = {1};
    for (auto& [p, e] : factors) {
        int sz = divisors.size();
        for (int i = 0; i < sz; i++) {
            i64 x = divisors[i];
            for (int j = 0; j < e; j++) {
                x *= p;
                divisors.push_back(x);
            }
        }
    }
    return divisors;
}
```

6 组合数学

6.1 组合数

```
template <typename T>
struct Combination {
    int n;
    vector<T> fact_, invfact_, inv_;

    Combination() : n(0), fact_{1}, invfact_{1}, inv_{0} {}
    Combination(int n_) : Combination() {
        init(n_);
    }

    void init(int n_) {
        if (n_ <= n) {
            return;
        }
        fact_.resize(n_ + 1);
        invfact_.resize(n_ + 1);
        inv_.resize(n_ + 1);

```

```
        for (int i = n + 1; i <= n_; i++) {
            fact_[i] = fact_[i - 1] * i;
        }
        invfact_[n_] = fact_[n_].inv();
        for (int i = n_; i > n; i--) {
            invfact_[i - 1] = invfact_[i] * i;
            inv_[i] = invfact_[i] * fact_[i - 1];
        }
        n = n_;
    }

    T fact(int k) {
        if (k > n) {
            init(2 * k);
        }
        return fact_[k];
    }

    T invfact(int k) {
        if (k > n) {
            init(2 * k);
        }
        return invfact_[k];
    }

    T inv(int k) {
        if (k > n) {
            init(2 * k);
        }
        return inv_[k];
    }

    T binom(int n, int m) {
        if (n < m || m < 0) {
            return 0;
        }
        return fact(n) * invfact(m) * invfact(n - m);
    }

    T perm(int n, int m) {
        if (n < m || m < 0) {
            return 0;
        }
        return fact(n) * invfact(n - m);
    }

    T multicomb(int n, int m) {
        if (n < 0 || m < 0) {
            return 0;
        }
        if (n == 0) {
            return (m == 0);
```

```
        }
        return binom(n + m - 1, m);
    }

    T Catalan(int n) {
        if (n < 0) {
            return 0;
        }
        return binom(2 * n, n) * inv(n + 1);
    }
};
```

6.2 康托展开

```
vector<int> toLehmer(vector<int>& p) {
    int n = p.size();
    Fenwick<AddMono<int>> fen(n);
    vector<int> c(n);
    for (int i = n - 1; i >= 0; i--) {
        c[i] = fen.query(p[i]);
        fen.add(p[i], 1);
    }
    return c;
}

vector<int> fromLehmer(vector<int>& c) {
    int n = c.size();
    vector<int> p(n);
    Fenwick<AddMono<int>> fen(n);
    for (int i = 0; i < n; i++) {
        fen.add(i, 1);
    }
    for (int i = 0; i < n; i++) {
        p[i] = fen.kth(c[i]);
        fen.add(p[i], -1);
    }
    return p;
}

i64 toRank(vector<int>& p) {
    int n = p.size();
    auto c = toLehmer(p);
    i64 fac = 1, rk = 0;
    for (int i = n - 1; i >= 0; i--) {
        rk += c[i] * fac;
        fac *= (n - i);
    }
    return rk;
}

```

```
vector<int> fromRank(int n, i64 rk) {
    vector<int> c(n);
    for (int i = 1; i <= n; i++) {
        c[n - i] = rk % i;
        rk /= i;
    }
    return fromLehmer(c);
}
```

6.3 逆序对

```
i64 invCount(vector<i64>& a, int l, int r, vector<i64>& tmp) {
    if (r - l <= 1) return 0;

    i64 res = 0;
    int mid = (l + r) / 2;
    res += invCount(a, l, mid, tmp);
    res += invCount(a, mid, r, tmp);

    int i = l, j = mid, k = l;
    while (i < mid && j < r) {
        if (a[i] <= a[j]) {
            tmp[k++] = a[i++];
        } else {
            tmp[k++] = a[j++];
            res += mid - i;
        }
    }
    while (i < mid) {
        tmp[k++] = a[i++];
    }
    while (j < r) {
        tmp[k++] = a[j++];
    }

    for (i = l; i < r; i++) {
        a[i] = tmp[i];
    }

    return res;
}

```

6.4 容斥原理

```
// cap(S) 表示 S 集合的交集大小

// 交集形式
```

```cpp
template <typename T, typename F>
T ie(int n, F&& cap) {
    T ans{};
    for (int S = 1; S < (1 << n); S++) {
        T cur = cap(S);
        if (popcount(u32(S)) & 1) {
            ans += cur;
        } else {
            ans -= cur;
        }
    }
    return ans;
}

// 补集形式
template <typename T, typename F>
T ieComp(int n, F&& cap) {
    T ans{};
    for (int S = 0; S < (1 << n); S++) {
        T cur = cap(S);
        if (popcount(u32(S)) & 1) {
            ans -= cur;
        } else {
            ans += cur;
        }
    }
    return ans;
}

// 恰好满足 k 个性质的元素数量
template <typename T, typename F>
vector<T> ieExact(int n, F&& cap) {
    vector<T> N(n + 1, T{});
    for (int S = 0; S < (1 << n); S++) {
        int k = popcount(u32(S));
        N[k] += cap(S);
    }

    vector<T> binom(n + 2, T{}), E(n + 1, T{});
    binom[0] = T{1};
    for (int j = 0; j <= n; j++) {
        for (int k = 0; k <= j; k++) {
            T term = N[j] * binom[k];
            if ((j - k) & 1) {
                E[k] -= term;
            } else {
                E[k] += term;
            }
        }
        for (int k = j + 1; k >= 1; k--) {
            binom[k] += binom[k - 1];
```

```
        }
    }
    return E;
}

// 至少满足 k 个
template <typename T, typename F>
vector<T> ieLeast(int n, F&& cap) {
    auto E = ieExact<T>(n, cap);
    for (int k = n - 1; k >= 0; k--) {
        E[k] += E[k + 1];
    }
    return E;
}

// 至多满足 k 个
template <typename T, typename F>
vector<T> ieMost(int n, F&& cap) {
    auto E = ieExact<T>(n, cap);
    for (int k = 0; k < n; k++) {
        E[k + 1] += E[k];
    }
    return E;
}
```

6.5 zeta 变换

```
// 子集和: f[mask] = sum_{sub ⊆ mask} f[sub]
// 方向: 小的 mask → 大的 mask（"吸收"不带的位）
template <typename T>
void subsetZeta(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask] += f[mask ^ i];
            }
        }
    }
}

// 子集和的逆
template <typename T>
void subsetMobius(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask] -= f[mask ^ i];
            }
```

```
        }
    }
}

// 超集和：f[mask] = sum_{sup ⊇ mask} f[sup]
// 方向：大的 mask → 小的 mask（"分发"给不带这一位的 mask）
template <typename T>
void supersetZeta(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask ^ i] += f[mask];
            }
        }
    }
}

// 超集和的逆
template <typename T>
void supersetMobius(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask ^ i] -= f[mask];
            }
        }
    }
}
```

6.6 Polyomino

```
struct Polyomino {
    using Cell = pair<int, int>;
    set<Cell> cells;
    Polyomino() {}
    Polyomino(const set<Cell>& cells_) : cells(cells_) {}
};

Polyomino normalize(const Polyomino& p) {
    if (p.cells.empty()) {
        return p;
    }
    int minX = p.cells.begin()->first;
    int minY = p.cells.begin()->second;
    for (auto& [x, y] : p.cells) {
        minX = min(minX, x);
        minY = min(minY, y);
    }
```

```
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(x - minX, y - minY);
    }
    return res;
}

Polyomino rotate(const Polyomino& p) {
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(y, -x);
    }
    return normalize(res);
}

Polyomino flip(const Polyomino& p) {
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(x, -y);
    }
    return normalize(res);
}
```

7 数据结构

7.1 线段树

7.1.1 线段树

```
template <typename Mono>
struct SegmentTree {
    using X = typename Mono::Type;

    int n;
    vector<X> info;

    SegmentTree() : n(0) {}
    SegmentTree(int n_) : SegmentTree(vector<X>(n_, Mono::e())) {}
    SegmentTree(const vector<X>& init) : n(init.size()) {
        int logn = __lg(n);
        info.assign(4 << logn, Mono::e());
        auto build = [&](auto&& self, int p, int l, int r) -> void {
            if (r - l == 1) {
                info[p] = init[l];
                return;
            }
            int mid = (l + r) / 2;
            self(self, 2 * p, l, mid);
            self(self, 2 * p + 1, mid, r);
            pull(p);
```

```
        };
        build(build, 1, 0, n);
    }

    void pull(int p) {
        info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
    }

    void modify(int p, int l, int r, int x, const X& v) {
        if (r - l == 1) {
            info[p] = v;
            return;
        }
        int mid = (l + r) / 2;
        if (x < mid) {
            modify(2 * p, l, mid, x, v);
        } else {
            modify(2 * p + 1, mid, r, x, v);
        }
        pull(p);
    }
    void modify(int p, const X& v) {
        modify(1, 0, n, p, v);
    }

    X query(int p, int l, int r, int x, int y) {
        if (y <= l || r <= x) {
            return Mono::e();
        }
        if (x <= l && r <= y) {
            return info[p];
        }
        int mid = (l + r) / 2;
        return Mono::op(query(2 * p, l, mid, x, y), query(2 * p + 1, mid, r, x, y));
    }
    X query(int l, int r) {
        return query(1, 0, n, l, r);
    }

    template <typename F>
    int findFirst(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        int res = findFirst(2 * p, l, mid, x, y, pred);
```

```
        if (res == -1) {
            res = findFirst(2 * p + 1, mid, r, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findFirst(int l, int r, F&& pred) {
        return findFirst(1, 0, n, l, r, pred);
    }

    template <typename F>
    int findLast(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        int res = findLast(2 * p + 1, mid, r, x, y, pred);
        if (res == -1) {
            res = findLast(2 * p, l, mid, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findLast(int l, int r, F&& pred) {
        return findLast(1, 0, n, l, r, pred);
    }

    template <typename F>
    int maxRight(int l, F&& pred, X& sum) {
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
    int minLeft(int r, F&& pred, X& sum) {
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
```

```
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.1.2 懒标记线段树

```
template <typename ActMono>
struct LazySegmentTree {
    using Mono = typename ActMono::Mono;
    using Tag = typename ActMono::Tag;
    using X = typename ActMono::X;
    using Y = typename ActMono::Y;

    int n;
    vector<X> info;
    vector<Y> tag;

    LazySegmentTree() : n(0) {}
    LazySegmentTree(int n_) : LazySegmentTree(vector<X>(n_, Mono::e())) {}
    LazySegmentTree(const vector<X>& init) : n(init.size()) {
        int logn = __lg(n);
        info.assign(4 << logn, Mono::e());
        tag.assign(4 << logn, Tag::e());
        auto build = [&](auto&& self, int p, int l, int r) -> void {
            if (r - l == 1) {
                info[p] = init[l];
                return;
            }
            int mid = (l + r) / 2;
            self(self, 2 * p, l, mid);
            self(self, 2 * p + 1, mid, r);
            pull(p);
        };
        build(build, 1, 0, n);
    }

    void pull(int p) {
        info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
    }

    void update(int p, int len, const Y& v) {
        info[p] = ActMono::act(info[p], v, len);
        tag[p] = Tag::op(tag[p], v);
    }

    void push(int p, int len) {
```

```
41          update(2 * p, len / 2, tag[p]);
42          update(2 * p + 1, (len + 1) / 2, tag[p]);
43          tag[p] = Tag::e();
44      }
45
46      void modify(int p, int l, int r, int x, const X& v) {
47          if (r - l == 1) {
48              info[p] = v;
49              return;
50          }
51          int mid = (l + r) / 2;
52          push(p, r - l);
53          if (x < mid) {
54              modify(2 * p, l, mid, x, v);
55          } else {
56              modify(2 * p + 1, mid, r, x, v);
57          }
58          pull(p);
59      }
60      void modify(int p, const X& v) {
61          modify(1, 0, n, p, v);
62      }
63
64      X query(int p, int l, int r, int x, int y) {
65          if (y <= l || r <= x) {
66              return Mono::e();
67          }
68          if (x <= l && r <= y) {
69              return info[p];
70          }
71          int mid = (l + r) / 2;
72          push(p, r - l);
73          return Mono::op(query(2 * p, l, mid, x, y), query(2 * p + 1, mid, r, x, y));
74      }
75      X query(int l, int r) {
76          return query(1, 0, n, l, r);
77      }
78
79      void apply(int p, int l, int r, int x, int y, const Y& v) {
80          if (y <= l || r <= x) {
81              return;
82          }
83          if (x <= l && r <= y) {
84              update(p, r - l, v);
85              return;
86          }
87          int mid = (l + r) / 2;
88          push(p, r - l);
89          apply(2 * p, l, mid, x, y, v);
90          apply(2 * p + 1, mid, r, x, y, v);
91          pull(p);
92      }
```

```
93      void apply(int l, int r, const Y& v) {
94          return apply(1, 0, n, l, r, v);
95      }
96
97      template <typename F>
98      int findFirst(int p, int l, int r, int x, int y, F&& pred) {
99          if (y <= l || r <= x) {
100             return -1;
101         }
102         if (x <= l && r <= y && !pred(info[p])) {
103             return -1;
104         }
105         if (r - l == 1) {
106             return l;
107         }
108         int mid = (l + r) / 2;
109         push(p, r - l);
110         int res = findFirst(2 * p, l, mid, x, y, pred);
111         if (res == -1) {
112             res = findFirst(2 * p + 1, mid, r, x, y, pred);
113         }
114         return res;
115     }
116     template <typename F>
117     int findFirst(int l, int r, F&& pred) {
118         return findFirst(1, 0, n, l, r, pred);
119     }
120
121     template <typename F>
122     int findLast(int p, int l, int r, int x, int y, F&& pred) {
123         if (y <= l || r <= x) {
124             return -1;
125         }
126         if (x <= l && r <= y && !pred(info[p])) {
127             return -1;
128         }
129         if (r - l == 1) {
130             return l;
131         }
132         int mid = (l + r) / 2;
133         push(p, r - l);
134         int res = findLast(2 * p + 1, mid, r, x, y, pred);
135         if (res == -1) {
136             res = findLast(2 * p, l, mid, x, y, pred);
137         }
138         return res;
139     }
140     template <typename F>
141     int findLast(int l, int r, F&& pred) {
142         return findLast(1, 0, n, l, r, pred);
143     }
144
```

```
    template <typename F>
    int maxRight(int l, F&& pred) {
        X sum = Mono::e();
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
    int minLeft(int r, F&& pred) {
        X sum = Mono::e();
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.1.3 动态开点线段树

```
template <typename ActMono>
struct DynamicSegmentTree {
    using Mono = typename ActMono::Mono;
    using Tag = typename ActMono::Tag;
    using X = typename ActMono::X;
    using Y = typename ActMono::Y;

    int n;
    int root;
    vector<X> info;
    vector<Y> tag;
    vector<int> lc, rc;

    DynamicSegmentTree() : n(0), root(-1) {}
    DynamicSegmentTree(int n_) : n(n_), root(-1) {
        root = newNode();
    }
    DynamicSegmentTree(const vector<X>& init) : n(init.size()), root(-1) {
        auto build = [&](auto&& self, int l, int r) -> int {
```

```
20              int p = newNode();
21              if (r - l == 1) {
22                  info[p] = init[l];
23                  return p;
24              }
25              int mid = (l + r) / 2;
26              lc[p] = self(self, l, mid);
27              rc[p] = self(self, mid, r);
28              pull(p);
29              return p;
30          };
31          root = build(build, 0, n);
32      }
33
34      int newNode() {
35          info.push_back(Mono::e());
36          tag.push_back(Tag::e());
37          lc.push_back(-1);
38          rc.push_back(-1);
39          return int(info.size()) - 1;
40      }
41
42      void pull(int p) {
43          info[p] = Mono::op(info[lc[p]], info[rc[p]]);
44      }
45
46      void update(int p, int len, const Y& v) {
47          info[p] = ActMono::act(info[p], v, len);
48          tag[p] = Tag::op(tag[p], v);
49      }
50
51      void push(int p, int len) {
52          if (lc[p] == -1) {
53              lc[p] = newNode();
54          }
55          if (rc[p] == -1) {
56              rc[p] = newNode();
57          }
58          update(lc[p], len / 2, tag[p]);
59          update(rc[p], (len + 1) / 2, tag[p]);
60          tag[p] = Tag::e();
61      }
62
63      void modify(int p, int l, int r, int x, const X& v) {
64          if (r - l == 1) {
65              info[p] = v;
66              return;
67          }
68          int mid = (l + r) / 2;
69          push(p, r - l);
70          if (x < mid) {
71              modify(lc[p], l, mid, x, v);
```

```
72          } else {
73              modify(rc[p], mid, r, x, v);
74          }
75          pull(p);
76      }
77      void modify(int x, const X& v) {
78          modify(root, 0, n, x, v);
79      }
80
81      X query(int p, int l, int r, int x, int y) {
82          if (y <= l || r <= x) {
83              return Mono::e();
84          }
85          if (x <= l && r <= y) {
86              return info[p];
87          }
88          int mid = (l + r) / 2;
89          push(p, r - l);
90          return Mono::op(query(lc[p], l, mid, x, y), query(rc[p], mid, r, x, y));
91      }
92      X query(int l, int r) {
93          return query(root, 0, n, l, r);
94      }
95
96      void apply(int p, int l, int r, int x, int y, const Y& v) {
97          if (y <= l || r <= x) {
98              return;
99          }
100         if (x <= l && r <= y) {
101             update(p, r - l, v);
102             return;
103         }
104         int mid = (l + r) / 2;
105         push(p, r - l);
106         apply(lc[p], l, mid, x, y, v);
107         apply(rc[p], mid, r, x, y, v);
108         pull(p);
109     }
110     void apply(int l, int r, const Y& v) {
111         apply(root, 0, n, l, r, v);
112     }
113
114     template <typename F>
115     int findFirst(int p, int l, int r, int x, int y, F&& pred) {
116         if (y <= l || r <= x) {
117             return -1;
118         }
119         if (x <= l && r <= y && !pred(info[p])) {
120             return -1;
121         }
122         if (r - l == 1) {
123             return l;
```

```
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findFirst(lc[p], l, mid, x, y, pred);
        if (res == -1) {
            res = findFirst(rc[p], mid, r, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findFirst(int l, int r, F&& pred) {
        return findFirst(root, 0, n, l, r, pred);
    }

    template <typename F>
    int findLast(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findLast(rc[p], mid, r, x, y, pred);
        if (res == -1) {
            res = findLast(lc[p], l, mid, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findLast(int l, int r, F&& pred) {
        return findLast(root, 0, n, l, r, pred);
    }

    template <typename F>
    int maxRight(int l, F&& pred, X& sum) {
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
```

```
    int minLeft(int r, F&& pred, X& sum) {
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.2 树状数组

7.2.1 树状数组

```
template <typename Mono>
struct Fenwick {
    using X = typename Mono::Type;

    int n;
    vector<X> a;

    Fenwick(int n_ = 0) : n(n_), a(n, Mono::e()) {}
    Fenwick(const vector<X>& init) : n(init.size()), a(init) {
        for (int i = 1; i <= n; i++) {
            int j = i + (i & -i);
            if (j <= n) {
                a[j - 1] = Mono::op(a[j - 1], a[i - 1]);
            }
        }
    }

    void add(int x, const X& v) {
        assert(0 <= x && x < n);
        for (int i = x + 1; i <= n; i += i & -i) {
            a[i - 1] = Mono::op(a[i - 1], v);
        }
    }

    X query(int x) {
        assert(0 <= x && x <= n);
        X res = Mono::e();
        for (int i = x; i > 0; i -= i & -i) {
            res = Mono::op(a[i - 1], res);
        }
        return res;
    }

```

```
    X query(int l, int r) {
        assert(l <= r);
        return Mono::op(Mono::inv(query(l)), query(r));
    }

    // lower_bound
    int kth(X k) {
        int x = 0;
        for (int i = 1 << __lg(n); i; i /= 2) {
            if (x + i <= n && k >= a[x + i - 1]) {
                x += i;
                k -= a[x - 1];
            }
        }
        return x;
    }
};
```

7.2.2 区间加、单点查树状数组

```
template <typename Abel>
struct DualFenwick {
    using X = typename Abel::Type;

    int n;
    Fenwick<Abel> fen;

    DualFenwick(int n_ = 0) : n(n_), fen(n) {}
    DualFenwick(const vector<X>& init) : n(init.size()) {
        vector<X> d(n);
        X pre = Abel::e();
        for (int i = 0; i < n; i++) {
            d[i] = Abel::op(Abel::inv(pre), init[i]);
            pre = init[i];
        }
        fen = Fenwick<Abel>(d);
    }

    void add(int l, int r, const X& v) {
        assert(0 <= l && l < r && r <= n);
        fen.add(l, v);
        if (r < n) {
            fen.add(r, Abel::inv(v));
        }
    }

    X query(int x) {
        assert(0 <= x && x < n);
        return fen.query(x + 1);
    }
```

```
};
```

7.2.3 区间加、区间和树状数组

```
template <typename Abel>
struct FenwickRangeAdd {
    using X = typename Abel::Type;

    int n;
    Fenwick<Abel> fen0, fen1;

    FenwickRangeAdd(int n_ = 0) : n(n_), fen0(n + 1), fen1(n + 1) {}
    FenwickRangeAdd(const vector<X>& init) : n(init.size()) {
        vector<X> d(n + 1), w(n + 1);
        X pre = Abel::e();
        for (int i = 0; i <= n; i++) {
            X cur = (i < n ? init[i] : Abel::e());
            d[i] = Abel::op(Abel::inv(pre), cur);
            w[i] = Abel::pow(d[i], -i);
            pre = cur;
        }
        fen0 = Fenwick<Abel>(w);
        fen1 = Fenwick<Abel>(d);
    }

    void add(int l, int r, const X& v) {
        assert(0 <= l && l < r && r <= n);
        fen0.add(l, Abel::pow(v, -l));
        fen0.add(r, Abel::pow(v, r));
        fen1.add(l, v);
        fen1.add(r, Abel::inv(v));
    }

    X query(int r) {
        assert(0 <= r && r <= n);
        return Abel::op(Abel::pow(fen1.query(r), r), fen0.query(r));
    }

    X query(int l, int r) {
        assert(0 <= l && l <= r && r <= n);
        return Abel::op(Abel::inv(query(l)), query(r));
    }
};
```

7.2.4 二维树状数组

```
template <typename Mono>
struct Fenwick2D {
    using X = typename Mono::Type;
```

```

    int h, w;
    vector<X> a;

    Fenwick2D(int h_ = 0, int w_ = 0) : h(h_), w(w_), a((h + 1) * (w + 1), Mono::e()) {}
    Fenwick2D(const vector<vector<X>>& init) : Fenwick2D(init.size(), init[0].size()) {
        for (int x = 0; x < h; x++) {
            assert(init[x].size() == w);
            for (int y = 0; y < w; y++) {
                a[(x + 1) * (w + 1) + (y + 1)] = init[x][y];
            }
        }
        for (int i = 1; i <= h; i++) {
            for (int j = 1; j <= w; j++) {
                int jj = j + (j & -j);
                if (jj <= w) {
                    a[i * (w + 1) + jj] = Mono::op(a[i * (w + 1) + jj], a[i * (w + 1) + j]);
                }
            }
        }
        for (int i = 1; i <= h; i++) {
            for (int j = 1; j <= w; j++) {
                int ii = i + (i & -i);
                if (ii <= h) {
                    a[ii * (w + 1) + j] = Mono::op(a[ii * (w + 1) + j], a[i * (w + 1) + j]);
                }
            }
        }
    }

    void add(int x, int y, const X& v) {
        assert(0 <= x && x < h);
        assert(0 <= y && y < w);
        for (int i = x + 1; i <= h; i += i & -i) {
            for (int j = y + 1; j <= w; j += j & -j) {
                a[i * (w + 1) + j] = Mono::op(a[i * (w + 1) + j], v);
            }
        }
    }

    X query(int x, int y) {
        assert(0 <= x && x <= h);
        assert(0 <= y && y <= w);
        X res = Mono::e();
        for (int i = x; i > 0; i -= i & -i) {
            for (int j = y; j > 0; j -= j & -j) {
                res = Mono::op(res, a[i * (w + 1) + j]);
            }
        }
        return res;
    }

```

```
    X query(int x1, int x2, int y1, int y2) {
        assert(0 <= x1 && x1 <= x2 && x2 <= h);
        assert(0 <= y1 && y1 <= y2 && y2 <= w);
        X s22 = query(x2, y2);
        X s12 = query(x1, y2);
        X s21 = query(x2, y1);
        X s11 = query(x1, y1);
        return Mono::op(Mono::op(s22, Mono::inv(s12)), Mono::op(Mono::inv(s21), s11));
    }
};
```

7.3 PBDS

7.3.1 哈希表

`HashMap<K, V, Hash> / HashSet<K, Hash>`。

`insert({k, v}) / insert(k)` 插入，返回 (迭代器，是否成功)；`erase(k)` 删除并返回是否存在；`find(k)` 查找，未找到返回 `end()`；`Map` 的 `a[k]` 访问值，键不存在时插入默认值。

上述操作期望 $O(1)$（插入、删除为均摊），最坏 $O(n)$；`clear()` 为 $O(n)$。

```
struct splitmix64Hash {
    static u64 splitmix64(u64 x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(u64 x) const {
        static const u64 FIXED_RANDOM = chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

template <typename K, typename V, typename Hash = splitmix64Hash>
using HashMap = __gnu_pbds::gp_hash_table<K, V, Hash>;

template <typename K, typename Hash = splitmix64Hash>
using HashSet = HashMap<K, __gnu_pbds::null_type, Hash>;
```

7.3.2 红黑树

`OrderedSet<K, Comp> / OrderedMap<K, V, Comp>`：键唯一，默认升序；下述大小关系按 `Comp`。

`insert` 插入，`erase(k)` 删除，`find(k)` 查找；`lower_bound(k) / upper_bound(k)` 返回首个 $\geq k$ / $> k$ 的迭代器；`Map` 的 `a[k]` 访问或插入默认值。均为 $O(\log n)$。

`find_by_order(k)` 返回从 0 计的第 `k` 个元素的迭代器，越界返回 `end()`；`order_of_key(k)` 返回严格排在 `k` 前的键数。均为 $O(\log n)$。

`a.split(k, b)`：`a` 保留 $\leq k$，$> k$ 移入 `b`，清空旧 `b`；$O(n+m)$，其中 m 为旧 `b` 大小。

`a.join(b)`：合并并清空 `b`，要求一棵树的所有键均小于另一棵树的所有键；$O(\log(n+m))$。

```cpp
template <typename K, typename V, typename Comp = less<K>>
using OrderedMap = __gnu_pbds::tree<
    K, V, Comp,
    __gnu_pbds::rb_tree_tag,
    __gnu_pbds::tree_order_statistics_node_update
>;

template <typename K, typename Comp = less<K>>
using OrderedSet = OrderedMap<K, __gnu_pbds::null_type, Comp>;
```

7.3.3 配对堆

`PairingHeap<T, Comp>`：默认大根堆，`greater<T>` 为小根堆。

`push(x)` 插入并返回句柄；`top()` 取堆顶；`pop()` 删除堆顶；`modify(it, x)` 修改句柄对应元素；`erase(it)` 删除元素。修改不使句柄失效，删除使其失效。

`a.join(b)` 合并并清空 `b`；`a.split(pred, b)` 将满足 `pred` 的元素移入 `b`，清空旧 `b`；`erase_if(pred)` 删除满足 `pred` 的元素并返回数量。

`top`、`push`、`join`：$O(1)$；`pop`、`modify`、`erase`：均摊 $O(\log n)$；`erase_if`、`clear`：$O(n)$；`split`：$O(n+m)$，m 为旧 `b` 大小。

```cpp
template <typename T, typename Comp = less<T>>
using PairingHeap = __gnu_pbds::priority_queue<T, Comp, __gnu_pbds::pairing_heap_tag>;
```

7.4 倍增

```cpp
template <typename Mono>
struct BinaryLifting {
    using X = typename Mono::Type;

    int n, logn;
    vector<vector<int>> nxt;
    vector<vector<X>> info;

    BinaryLifting() : n(0) {}
    BinaryLifting(int n_) : n(n_), logn(__lg(n)) {
        nxt.assign(logn + 1, vector<int>(n, -1));
        info.assign(logn + 1, vector<X>(n, Mono::e()));
    }

    void add(int u, int v, const X& x) {
        assert(-1 <= v && v < n);
        nxt[0][u] = v;
        info[0][u] = x;
    }

    void build() {
        for (int k = 0; k < logn; k++) {
            for (int u = 0; u < n; u++) {
                int v = nxt[k][u];
```

```
                if (v == -1) {
                    nxt[k + 1][u] = -1;
                    info[k + 1][u] = info[k][u];
                } else {
                    nxt[k + 1][u] = nxt[k][v];
                    info[k + 1][u] = Mono::op(info[k][u], info[k][v]);
                }
            }
        }
    }

    pair<int, X> query(int u, i64 step) {
        assert(0 <= step && step < (1LL << (logn + 1)));
        X res = Mono::e();
        for (int k = logn; k >= 0 && u != -1; k--) {
            if ((step >> k) & 1) {
                res = Mono::op(res, info[k][u]);
                u = nxt[k][u];
            }
        }
        return {u, res};
    }

    int jump(int u, i64 step) {
        assert(0 <= step && step < (1LL << (logn + 1)));
        for (int k = logn; k >= 0 && u != -1; k--) {
            if ((step >> k) & 1) {
                u = nxt[k][u];
            }
        }
        return u;
    }
};
```

7.5 并查集

```
struct DSU {
    vector<int> f, siz;

    DSU() {}
    DSU(int n) {
        f.resize(n);
        iota(f.begin(), f.end(), 0);
        siz.assign(n, 1);
    }

    int find(int x) {
        while (x != f[x]) {
            x = f[x] = f[f[x]];
        }
```

```
        return x;
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool merge(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            return false;
        }
        siz[x] += siz[y];
        f[y] = x;
        return true;
    }

    int size(int x) {
        return siz[find(x)];
    }
};
```

7.6 稀疏表

```
template <typename Mono>
struct SparseTable {
    using X = typename Mono::Type;

    int n;
    vector<vector<X>> st;

    SparseTable() : n(0) {}
    SparseTable(const vector<X>& a) : n(a.size()) {
        int logn = __lg(n);
        st.assign(logn + 1, vector<X>(n));
        for (int i = 0; i < n; i++) {
            st[0][i] = a[i];
        }
        for (int j = 1; (1 << j) <= n; j++) {
            for (int i = 0; i + (1 << j) <= n; i++) {
                st[j][i] = Mono::op(st[j - 1][i], st[j - 1][i + (1 << (j - 1))]);
            }
        }
    }

    X query(int l, int r) {
        assert(l < r);
        int len = r - l;
        int t = __lg(len);
```

```
        return Mono::op(st[t][l], st[t][r - (1 << t)]);
    }
};
```

7.7 01-Trie

```
struct BinaryTrie {
    struct Node {
        array<int, 2> ch;
        int end;
        Node() {
            ch[0] = ch[1] = 0;
            end = 0;
        }
    };

    vector<Node> tr;
    int tot;

    BinaryTrie(int n = 0) : tr(n), tot(1) {}

    void insert(int x) {
        int p = 1;
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
            if (!tr[p].ch[c]) {
                ++tot;
                tr[p].ch[c] = tot;
            }
            p = tr[p].ch[c];
            tr[p].end++;
        }
    }

    int query(int x) {
        int p = 1, res = 0;
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
            if (tr[p].ch[c ^ 1]) {
                res |= (1 << i);
                p = tr[p].ch[c ^ 1];
            } else {
                p = tr[p].ch[c];
            }
        }
        return res;
    }

    void erase(int x) {
        int p = 1;
```

```
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
            int nxt = tr[p].ch[c];
            tr[nxt].end--;
            if (tr[nxt].end == 0) {
                tr[p].ch[c] = 0;
            }
            p = nxt;
        }
    }
};

```

7.8 笛卡尔树

```
template <typename T, bool IS_MIN = false>
struct CartesianTree {
    int n;
    vector<T>& a;
    vector<int> L, R, p;
    // range[i]: the interval of the subtree rooted at i
    vector<pair<int, int>> range;
    int root;

    CartesianTree(vector<T>& a_) : n(a_.size()), a(a_), root(-1) {
        L.assign(n, -1);
        R.assign(n, -1);
        p.assign(n, -1);
        range.assign(n, {0, n});
        vector<int> stk;

        auto cmp = [&](int i, int j) -> bool {
            if (a[i] == a[j]) {
                return i < j;
            }
            if constexpr (IS_MIN) {
                return a[i] < a[j];
            } else {
                return a[j] < a[i];
            }
        };

        for (int i = 0; i < n; i++) {
            while (!stk.empty() && cmp(i, stk.back())) {
                L[i] = stk.back();
                stk.pop_back();
            }
            range[i].first = (stk.empty() ? 0 : stk.back() + 1);
            if (!stk.empty()) {
                R[stk.back()] = i;
```

```
            }
            stk.push_back(i);
        }
        if (!stk.empty()) {
            root = stk.front();
        }
        stk.clear();
        for (int i = n - 1; i >= 0; i--) {
            while (!stk.empty() && cmp(i, stk.back())) {
                stk.pop_back();
            }
            range[i].second = stk.empty() ? n : stk.back();
            stk.push_back(i);
        }
        for (int i = 0; i < n; i++) {
            if (L[i] != -1) {
                p[L[i]] = i;
            }
            if (R[i] != -1) {
                p[R[i]] = i;
            }
        }
    }
};
```

8 线性代数

8.1 矩阵

```
template <typename T>
struct Matrix {
    int n, m;
    vector<vector<T>> a;

    Matrix() : n(0), m(0) {}
    Matrix(int n_, int m_, T v = T{}) : n(n_), m(m_), a(n, vector<T>(m, v)) {}
    Matrix(int n_) : Matrix(n_, n_) {}

    static Matrix identity(int n) {
        Matrix res(n);
        for (int i = 0; i < n; i++) {
            res[i][i] = T{1};
        }
        return res;
    }

    vector<T>& operator[](int i) {
        return a[i];
    }
    const vector<T>& operator[](int i) const {
```

```
22          return a[i];
23      }
24  };
25
26  template <typename T>
27  Matrix<T> operator*(const Matrix<T>& a, const Matrix<T>& b) {
28      assert(a.m == b.n);
29      Matrix<T> c(a.n, b.m);
30      for (int i = 0; i < c.n; i++) {
31          for (int k = 0; k < a.m; k++) {
32              for (int j = 0; j < c.m; j++) {
33                  c.a[i][j] += a.a[i][k] * b.a[k][j];
34              }
35          }
36      }
37      return c;
38  }
39
40  template <typename T>
41  Matrix<T>& operator*=(Matrix<T>& a, const Matrix<T>& b) {
42      return a = a * b;
43  }
44
45  template <typename T, integral U>
46  Matrix<T> power(Matrix<T> a, U b) {
47      Matrix<T> res = Matrix<T>::identity(a.n);
48      for (; b > 0; b >>= 1) {
49          if (b & 1) {
50              res *= a;
51          }
52          a *= a;
53      }
54      return res;
55  }
56
57  template <typename T>
58  Matrix<T> transpose(const Matrix<T>& a) {
59      Matrix<T> res(a.m, a.n);
60      for (int i = 0; i < a.n; i++) {
61          for (int j = 0; j < a.m; j++) {
62              res[j][i] = a[i][j];
63          }
64      }
65      return res;
66  }
67
68  template <typename T>
69  Matrix<T> augment(const Matrix<T>& a, const Matrix<T>& b) {
70      assert(a.n == b.n);
71      Matrix<T> res(a.n, a.m + b.m);
72      for (int i = 0; i < a.n; i++) {
73          for (int j = 0; j < a.m; j++) {
```

```
            res[i][j] = a[i][j];
        }
        for (int j = 0; j < b.m; j++) {
            res[i][a.m + j] = b[i][j];
        }
    }
    return res;
}

template <typename T>
Matrix<T> augment(const Matrix<T>& a, const vector<T>& b) {
    assert(a.n == int(b.size()));
    Matrix<T> res(a.n, a.m + 1);
    for (int i = 0; i < a.n; i++) {
        for (int j = 0; j < a.m; j++) {
            res[i][j] = a[i][j];
        }
        res[i][a.m] = b[i];
    }
    return res;
}
```

8.2 高斯消元

```
template <typename T>
pair<vector<int>, T> eliminate(Matrix<T>& A, int lim, bool full) {
    // full == true -> RREF
    // full == false -> REF
    vector<int> pivot;
    T prod = T{1};
    int r = 0;
    for (int j = 0; j < lim && r < A.n; j++) {
        int p = r;
        while (p < A.n && A[p][j] == T{0}) {
            p++;
        }
        if (p == A.n) {
            continue;
        }
        if (p != r) {
            swap(A[p], A[r]);
            prod = -prod;
        }
        T x = A[r][j];
        prod *= x;
        pivot.push_back(j);
        T inv = T{1} / x;
        for (int k = j; k < A.m; k++) {
            A[r][k] *= inv;
        }
```

```
        for (int i = 0; i < A.n; i++) {
            if (i == r || (!full && i < r)) {
                continue;
            }
            T c = A[i][j];
            if (c == T{0}) {
                continue;
            }
            A[i][j] = T{0};
            for (int k = j + 1; k < A.m; k++) {
                A[i][k] -= c * A[r][k];
            }
        }
        r++;
    }
    return {pivot, prod};
}

template <typename T>
pair<vector<int>, T> gauss(Matrix<T>& A) {
    return eliminate(A, A.m, false);
}
template <typename T>
pair<vector<int>, T> gaussJordan(Matrix<T>& A, int lim) {
    return eliminate(A, lim, true);
}

template <typename T>
pair<T, Matrix<T>> matrixInv(const Matrix<T>& a) {
    if (a.n != a.m) {
        return {T{0}, {}};
    }
    int n = a.n;
    auto aug = augment(a, Matrix<T>::identity(n));
    auto [pivot, det] = gaussJordan(aug, n);
    if (pivot.size() < n) {
        return {T{0}, {}};
    }
    Matrix<T> inv(n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            inv[i][j] = aug[i][n + j];
        }
    }
    return {det, inv};
}

template <typename T>
int matrixRank(Matrix<T> a) {
    auto [pivot, prod] = gauss(a);
    return pivot.size();
}
```

```cpp

template <typename T>
T determinant(Matrix<T> A) {
    assert(A.n == A.m);
    auto [pivot, det] = gauss(A);
    if (pivot.size() < A.n) {
        return T{0};
    }
    return det;
}

template <typename T>
vector<vector<T>> solveLinear(const Matrix<T>& A, const vector<T>& b) {
    int n = A.n;
    int m = A.m;
    auto aug = augment(A, b);
    auto [pivot, prod] = gaussJordan(aug, m);
    int r = pivot.size();
    for (int i = r; i < n; i++) {
        if (aug[i][m] != T{0}) {
            return {};
        }
    }
    vector<int> row(m, -1);
    for (int i = 0; i < r; i++) {
        row[pivot[i]] = i;
    }
    vector<vector<T>> sol(1, vector<T>(m));
    for (int j = 0; j < m; j++) {
        if (row[j] != -1) {
            sol[0][j] = aug[row[j]][m];
        }
    }
    for (int j = 0; j < m; j++) {
        if (row[j] != -1) {
            continue;
        }
        vector<T> x(m);
        x[j] = T{-1};
        for (int i = 0; i < r; i++) {
            x[pivot[i]] = aug[i][j];
        }
        sol.push_back(x);
    }
    return sol;
}
```

8.3 GF(2) 空间

8.3.1 异或线性基

```cpp
struct Basis {
    array<u64, 64> p{};
    bool norm = false;

    int size() {
        int r = 0;
        for (int i = 0; i < 64; i++) {
            r += p[i] != 0;
        }
        return r;
    }

    bool insert(u64 x) {
        for (int i = 63; i >= 0; i--) {
            if (!(x >> i & 1)) {
                continue;
            }
            if (!p[i]) {
                p[i] = x;
                norm = false;
                return true;
            }
            x ^= p[i];
        }
        return false;
    }

    bool contain(u64 x) {
        for (int i = 63; i >= 0; i--) {
            if (x >> i & 1) {
                x ^= p[i];
            }
        }
        return x == 0;
    }

    u64 getMax(u64 x = 0) {
        for (int i = 63; i >= 0; i--) {
            if ((x ^ p[i]) > x) {
                x ^= p[i];
            }
        }
        return x;
    }

    u64 getMin(u64 x = 0) {
        for (int i = 63; i >= 0; i--) {
            if ((x ^ p[i]) < x) {
```

```
                x ^= p[i];
            }
        }
        return x;
    }

    void normalize() {
        if (norm == true) {
            return;
        }
        norm = true;
        for (int i = 63; i >= 0; i--) {
            if (!p[i]) {
                continue;
            }
            for (int j = i - 1; j >= 0; j--) {
                if (p[i] >> j & 1) {
                    p[i] ^= p[j];
                }
            }
        }
    }

    u64 kth(u64 k) {
        normalize();
        vector<u64> b;
        for (int i = 0; i < 64; i++) {
            if (p[i]) {
                b.push_back(p[i]);
            }
        }
        int r = b.size();
        if (r < 64) {
            if (k >= (1ULL << r)) {
                return -1;
            }
        }
        u64 res = 0;
        for (int i = 0; i < r; i++) {
            if (k >> i & 1) {
                res ^= b[i];
            }
        }
        return res;
    }
};
```

8.3.2 逆矩阵

```
namespace gf2 {
```

```

template <typename U>
vector<U> inverse(int n, vector<U> a) {
    vector<U> b(n);
    for (int i = 0; i < n; i++) {
        b[i] = U{1} << i;
    }
    for (int j = 0; j < n; j++) {
        int p = j;
        while (p < n && !(a[p] >> j & U{1})) {
            p++;
        }
        if (p == n) {
            return {};
        }
        swap(a[p], a[j]);
        swap(b[p], b[j]);
        for (int i = 0; i < n; i++) {
            if (i != j && (a[i] >> j & U{1})) {
                a[i] ^= a[j];
                b[i] ^= b[j];
            }
        }
    }
    return b;
}

}
```

8.3.3 转置

```
namespace gf2 {

template <typename U>
vector<U> transpose(int n, int m, const vector<U>& a) {
    vector<U> b(m);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (a[i] >> j & U{1}) {
                b[j] |= U{1} << i;
            }
        }
    }
    return b;
}

}
```

8.3.4 GF(2) 线性方程组

```cpp
namespace gf2 {

template <unsigned_integral U>
vector<U> solveLinear(int n, int m, vector<U> a, U b) {
    vector<U> mat(m);
    U rhs = 0;
    for (int i = 0; i < n; i++) {
        U x = a[i], c = b >> i & 1;
        for (int j = m - 1; j >= 0; j--) {
            if ((x ^ mat[j]) < x) {
                x ^= mat[j];
                c ^= rhs >> j & 1;
            }
        }
        if (!x) {
            if (c) {
                return {};
            }
            continue;
        }
        int k = __lg(x);
        for (int j = k; j < m; j++) {
            if ((mat[j] ^ x) < mat[j]) {
                mat[j] ^= x;
                rhs ^= c << j;
            }
        }
        mat[k] = x;
        rhs |= c << k;
    }
    vector<U> res{rhs};
    mat = transpose(m, m, mat);
    for (int j = 0; j < m; j++) {
        if (!(mat[j] >> j & 1)) {
            res.push_back(mat[j] | (U{1} << j));
        }
    }
    return res;
}

}
```

9 字符串

9.1 Manacher

```cpp
vector<int> manacher(string s) {
    string t = "#";
```

```
    for (auto c : s) {
        t += c;
        t += '#';
    }
    int n = t.size();
    vector<int> r(n);
    for (int i = 0, j = 0; i < n; i++) {
        if (2 * j - i >= 0 && j + r[j] > i) {
            r[i] = min(r[2 * j - i], j + r[j] - i);
        }
        while (i - r[i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]]) {
            r[i] += 1;
        }
        if (i + r[i] > j + r[j]) {
            j = i;
        }
    }
    return r;
}
```

9.2 KMP

```
vector<int> kmp(string s) {
    int n = s.size();
    vector<int> f(n + 1);
    for (int i = 1, j = 0; i < n; i++) {
        while (j && s[i] != s[j]) {
            j = f[j];
        }
        j += (s[i] == s[j]);
        f[i + 1] = j;
    }
    return f;
}

```

9.3 Trie

```
struct Trie {
    struct Node {
        array<int, 26> nxt;
        int pass, end;
        Node() {
            nxt.fill(-1);
            end = 0;
            pass = 0;
        }
    };

```

```
    vector<Node> tr;
    int tot;

    Trie(int n) {
        init(n);
    }

    void init(int n) {
        tr.assign(n, {});
        tot = 0;
    }

    void insert(string& s) {
        int p = 0;
        ++tr[p].pass;
        for (char& ch : s) {
            int c = ch - 'a';
            if (tr[p].nxt[c] == -1) {
                ++tot;
                tr[p].nxt[c] = tot;
            }
            p = tr[p].nxt[c];
            ++tr[p].pass;
        }
        ++tr[p].end;
    }

    int count(string& s) {
        int p = 0;
        for (char& ch : s) {
            int c = ch - 'a';
            if (tr[p].nxt[c] == -1) {
                return false;
            }
            p = tr[p].nxt[c];
        }
        return tr[p].end;
    }
};

```

9.4 Z 函数

```
vector<int> Z(string s) {
    int n = s.size();
    vector<int> z(n + 1);
    z[0] = n;
    for (int i = 1, j = 1; i < n; i++) {
        z[i] = max(0, min(j + z[j] - i, z[i - j]));
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
```

```
            z[i]++;
        }
        if (i + z[i] > j + z[j]) {
            j = i;
        }
    }
    return z;
}

```

9.5 AC 自动机

```
struct AhoCorasick {
    static constexpr int ALPHABET = 26;
    struct Node {
        int len;
        int link;
        array<int, ALPHABET> next;
        Node() : len{0}, link{0}, next{} {}
    };

    vector<Node> t;

    AhoCorasick() {
        init();
    }

    void init() {
        t.assign(2, Node());
        t[0].next.fill(1);
        t[0].len = -1;
    }

    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }

    int add(const string &a) {
        int p = 1;
        for (auto c : a) {
            int x = c - 'a';
            if (t[p].next[x] == 0) {
                t[p].next[x] = newNode();
                t[t[p].next[x]].len = t[p].len + 1;
            }
            p = t[p].next[x];
        }
        return p;
    }
```

```

    void work() {
        queue<int> q;
        q.push(1);

        while (!q.empty()) {
            int x = q.front();
            q.pop();

            for (int i = 0; i < ALPHABET; i++) {
                if (t[x].next[i] == 0) {
                    t[x].next[i] = t[t[x].link].next[i];
                } else {
                    t[t[x].next[i]].link = t[t[x].link].next[i];
                    q.push(t[x].next[i]);
                }
            }
        }
    }

    int next(int p, int x) {
        return t[p].next[x];
    }

    int link(int p) {
        return t[p].link;
    }

    int len(int p) {
        return t[p].len;
    }

    int size() {
        return t.size();
    }
};

```

10 随机化

10.1 随机数

```
static mt19937_64 rng(static_cast<u64>(chrono::steady_clock::now().time_since_epoch().count()));

i64 randInt(i64 l, i64 r) {
    assert(l < r);
    return uniform_int_distribution<i64>(l, r - 1)(rng);
}

u64 randInt(u64 n) {
```

```
    assert(n > 0);
    return uniform_int_distribution<u64>(0, n - 1)(rng);
}

bool randBool() {
    return randInt(2);
}

double randReal(double lo, double hi) {
    assert(lo < hi);
    return uniform_real_distribution<double>(lo, hi)(rng);
}
```

10.2 随机数组

```
template <typename T>
void shuffle(vector<T>& a) {
    int n = a.size();
    for (int i = 0; i < n; i++) {
        int j = randInt(0, i + 1);
        if (i != j) {
            swap(a[i], a[j]);
        }
    }
}
// OR ranges::shuffle(a, rng);

template <typename T>
T& randElem(vector<T>& a) {
    assert(!a.empty());
    return a[randInt(a.size())];
}

vector<i64> randArray(int n, i64 l, i64 r) {
    vector<i64> a(n);
    for (auto& x : a) {
        x = randInt(l, r);
    }
    return a;
}

vector<int> randPerm(int n) {
    vector<int> a(n);
    iota(a.begin(), a.end(), 0);
    shuffle(a);
    return a;
}

vector<i64> randDistinct(int n, i64 l, i64 r) {
    unordered_set<i64> s;
```

```
    vector<i64> a;
    for (i64 i = r - l - n; i < r - l; i++) {
        i64 x = randInt(i + 1);
        if (!s.insert(x).second) {
            s.insert(i), a.push_back(i + l);
        } else {
            a.push_back(x + l);
        }
    }
    shuffle(a);
    return a;
}
```

10.3 随机图

```
vector<pair<int, int>> randGraph(int n, int m) {
    set<pair<int, int>> s;
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u != v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}

vector<pair<int, int>> randTree(int n) {
    auto p = randPerm(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) {
        e.emplace_back(p[randInt(i)], p[i]);
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randChain(int n) {
    auto p = randPerm(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) {
        e.emplace_back(p[i - 1], p[i]);
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randStar(int n) {
    auto p = randPerm(n);
```

```
36      vector<pair<int, int>> e;
37      for (int i = 1; i < n; i++) {
38          e.emplace_back(p[0], p[i]);
39      }
40      shuffle(e);
41      return e;
42  }
43
44  vector<pair<int, int>> randConnectedGraph(int n, int m) {
45      set<pair<int, int>> s;
46      for (auto [u, v] : randTree(n)) {
47          s.emplace(min(u, v), max(u, v));
48      }
49      while (m > int(s.size())) {
50          int u = randInt(n), v = randInt(n);
51          if (u > v) {
52              swap(u, v);
53          }
54          if (u != v) {
55              s.emplace(u, v);
56          }
57      }
58      vector<pair<int, int>> e(s.begin(), s.end());
59      for (auto& [u, v] : e) {
60          if (randBool()) {
61              swap(u, v);
62          }
63      }
64      shuffle(e);
65      return e;
66  }
67
68  vector<pair<int, int>> randDAG(int n, int m) {
69      auto p = randPerm(n);
70      set<pair<int, int>> s;
71      while (m > int(s.size())) {
72          int u = randInt(n), v = randInt(n);
73          if (u < v) {
74              s.emplace(u, v);
75          }
76      }
77      vector<pair<int, int>> e;
78      for (auto [u, v] : s) {
79          e.emplace_back(p[u], p[v]);
80      }
81      shuffle(e);
82      return e;
83  }
84
85  vector<pair<int, int>> randBipartiteGraph(int n, int k, int m) {
86      set<pair<int, int>> s;
87      while (m > int(s.size())) {
```

```
        s.emplace(randInt(n), n + randInt(k));
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}

vector<pair<int, int>> randSCC(int n, int m) {
    auto p = randPerm(n);
    set<pair<int, int>> s;
    if (n > 1)
        for (int i = 0; i < n; i++)
            s.emplace(p[i], p[(i + 1) % n]);
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u != v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}
```

10.4 模拟退火

```
// f 越大越好，g(cur, t) 返回新状态，可用温度 t 控制扰动。
template <typename T, typename F, typename G>
T simulatedAnnealing(T cur, F&& f, G&& g, double t0, double t1, double sec) {
    assert(t0 > t1 && t1 > 0 && sec > 0);
    auto beg = chrono::steady_clock::now();
    T ans = cur;
    auto val = f(cur), res = val;
    while (true) {
        double p = chrono::duration<double>(chrono::steady_clock::now() - beg).count() / sec;
        if (p >= 1) {
            break;
        }
        double t = exp(lerp(log(t0), log(t1), p));
        T nxt = g(cur, t);
        auto v = f(nxt);
        double d = double(v) - double(val);
        if (d >= 0 || randReal(0, 1) < exp(d / t)) {
            cur = move(nxt);
            val = v;
        }
        if (val > res) {
            ans = cur;
            res = val;
        }
```

```
    }
    return ans;
}
```

11 杂项

11.1 二分答案

```
template <typename T, typename F>
T firstTrue(T lo, T hi, F&& pred) {
    while (lo < hi) {
        T mid = midpoint(lo, hi);
        if (pred(mid)) {
            hi = mid;
        } else {
            lo = mid + 1;
        }
    }
    return lo;
}

template <typename T, typename F>
T lastTrue(T lo, T hi, F&& pred) {
    while (lo < hi) {
        T mid = midpoint(hi, lo);
        if (pred(mid)) {
            lo = mid;
        } else {
            hi = mid - 1;
        }
    }
    return lo;
}
```

11.2 三分

```
template <typename T, typename F, typename C = less<>>
T ternarySearch(T lo, T hi, F&& f, C&& comp = {}) {
    for (int i = 0; i < 100; i++) {
        T m1 = lo + (hi - lo) / 3;
        T m2 = hi - (hi - lo) / 3;
        if (comp(f(m1), f(m2))) {
            hi = m2;
        } else {
            lo = m1;
        }
    }
    return (lo + hi) / 2;
```

```
}
```

11.3 i128

```
using i128 = __int128;

ostream& operator<<(ostream& os, i128 n) {
    if (n == 0) {
        return os << 0;
    }
    bool neg = n < 0;
    u128 x = neg ? u128(0) - u128(n) : u128(n);
    string s;
    while (x > 0) {
        s += char('0' + x % 10);
        x /= 10;
    }
    if (neg) s += '-';
    reverse(s.begin(), s.end());
    return os << s;
}

i128 toi128(const string& s) {
    bool neg = !s.empty() && s[0] == '-';
    int start = !s.empty() && (s[0] == '-' || s[0] == '+');
    i128 n = 0;
    for (int i = start; i < int(s.size()); i++) {
        int d = s[i] - '0';
        n = neg ? n * 10 - d : n * 10 + d;
    }
    return n;
}

i128 sqrti128(i128 n) {
    assert(n >= 0);
    i128 lo = 0, hi = i128(1) << 64;
    while (lo < hi) {
        i128 x = (lo + hi + 1) / 2;
        if (x <= n / x) {
            lo = x;
        } else {
            hi = x - 1;
        }
    }
    return lo;
}

i128 gcd(i128 a, i128 b) {
    while (b) {
        a %= b;
```

```
        swap(a, b);
    }
    return a;
}
```

11.4 日期

```
const int d[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

bool isLeap(int y) {
    return y % 400 == 0 || (y % 4 == 0 && y % 100 != 0);
}

int daysInMonth(int y, int m) {
    return d[m - 1] + (isLeap(y) && m == 2);
}

int getDay(int y, int m, int d) {
    int ans = 0;
    for (int i = 1970; i < y; i++) {
        ans += 365 + isLeap(i);
    }
    for (int i = 1; i < m; i++) {
        ans += daysInMonth(y, i);
    }
    ans += d;
    return (ans + 2) % 7 + 1;
}

```

11.5 template

```
using namespace std;

using i64 = int64_t;
using u64 = uint64_t;
using u32 = uint32_t;

using i128 = __int128_t;
using u128 = __uint128_t;

template <typename T>
constexpr T inf = numeric_limits<T>::max() / 2;
template <>
constexpr int inf<int> = 1e9;
template <>
constexpr i64 inf<i64> = 1e18;

void solve();
```

```

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t;
    cin >> t;

    while (t--) {
        solve();
    }

    return 0;
}
```

11.6 debug

```
using namespace std;

#define dbg(...) do { cerr << "[" << __LINE__ << "] "; pr(__VA_ARGS__); cerr << '\n'; } while (0)

template <class T, class... U>
void pr(T x, U... y) {
    if constexpr (requires { cerr << x; }) {
        cerr << x;
    } else if constexpr (requires { tuple_size<T>::value; }) {
        cerr << '(';
        bool f = 0;
        apply([&](auto... e) { ((cerr << (exchange(f, 1) ? "," : ""), pr(e)), ...); }, x);
        cerr << ')';
    } else {
        cerr << '[';
        bool f = 0;
        for (auto e : x) cerr << (exchange(f, 1) ? "," : ""), pr(e);
        cerr << ']';
    }
    if constexpr (sizeof...(y)) cerr << "  ", pr(y...);
}
```

11.7 对拍

```
#!/bin/bash
# bash stress.sh ./sol ./brute ./gen
set -e
for ((i = 1; ; i++)); do
    "$3" > in.txt
    "$1" < in.txt > out.txt
    "$2" < in.txt > ans.txt
```

```
    if ! diff -w out.txt ans.txt > diff.txt; then
        printf '\nWA %d\n' "$i"
        break
    fi
    printf '\rAC %d' "$i"
done
```

11.8 对拍(with SPJ)

```
#!/bin/bash
# bash stress_spj.sh ./sol ./spj ./gen
set -e
for ((i = 1; ; i++)); do
    "$3" > in.txt
    "$1" < in.txt > out.txt
    if ! "$2" in.txt out.txt; then
        printf '\nWA %d\n' "$i"
        break
    fi
    printf '\rAC %d' "$i"
done
```

11.9 SPJ

```
using namespace std;

int main(int argc, char* argv[]) {
    ifstream in(argv[1]), out(argv[2]);
    return 0;
}
```